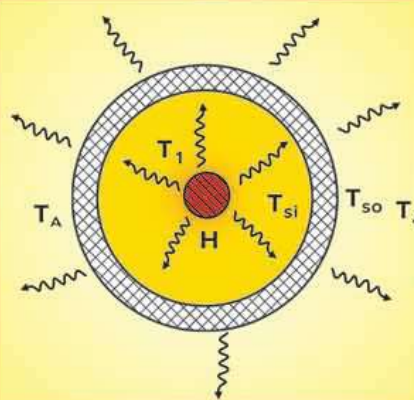
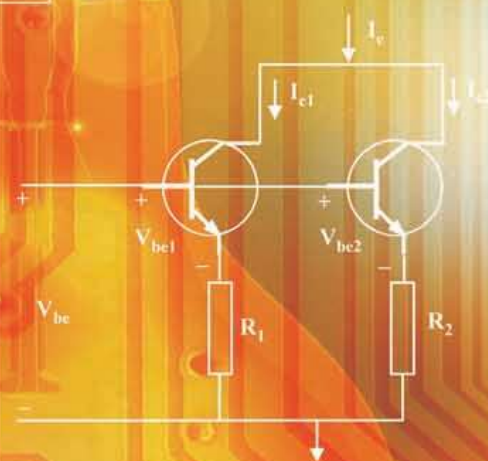
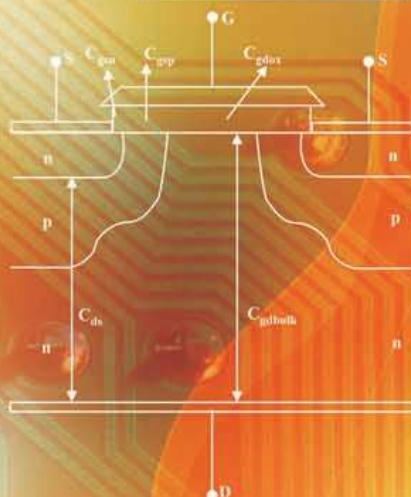
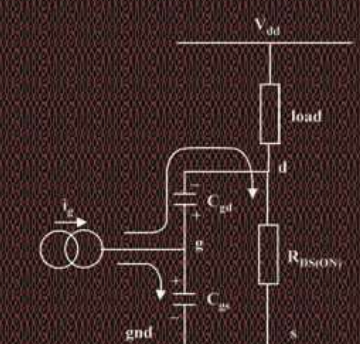
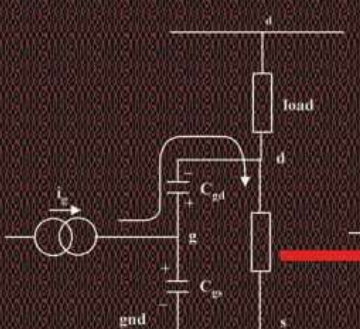
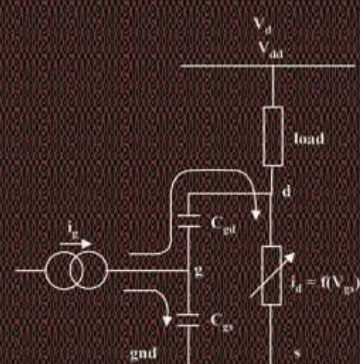
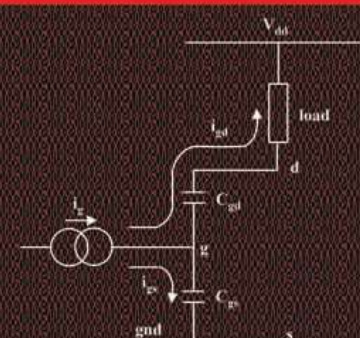




Power Electronics

Essentials and Applications

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Foreword

The book that you have in your hands is the outcome of years of patient but enthusiastic learning and exploration of the field of power electronics by its author. You may feel a bit apprehensive in coming to terms with more than eight hundred pages on this topic; however, a closer look at the table of contents will reveal the comprehensive coverage of the subject, explored from many angles and with various tools. This book, indeed, reflects well the professional trajectory of its author. Having known Dr L. Umanand from his student days and later as a colleague at the Centre for Electronics Design and Technology, I had the pleasure to interact with him during his entire professional career and share his passion for power electronics. I can clearly recognize in this book that same passion that he progressively developed for various aspects of the subject.

He has taken up the challenge to present here a comprehensive synthesis of the field with the required theoretical approach, combined with the pragmatism that comes from experience accumulated through years of direct involvement in design, development and testing of the power electronics systems. This, together with the originality of certain topics like Bond graphs, Design of Magnetism or Design for Reliability, will surely make this book stand out.

The book reflects equally well the deep interest the author has taken in the last few years of the “teaching/learning” process. The learning objectives of each chapter are clearly stated and the material is illustrated with numerous examples. The reader is also challenged throughout the book with numerous questions, problems, laboratory assignments, that, if carried out sincerely, should ensure proper anchoring of the newly acquired knowledge.

In spite of the limitations of any written material, I wish that through this book Dr Umanand will be able to inspire you in sharing his passion and enthusiasm for the subject as he does so well with his students, even if you miss the immediate proximity of his broad smile and great sense of humor.

Andre Pittet

Chief Project Advisor
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Preface

The subject of power electronics has been treated in numerous literatures with various viewpoints and styles. At the outset, it may appear as though one more book is being introduced into the market, which may at most be a marginal update on the existing literature with the same topics that are addressed in other books. Likewise, I would like to reiterate that the topics and concepts covered in this book are not new as one would expect, considering the field of power electronics is rather mature. Notwithstanding the maturity of the field, I must add that the topics in this book are neither old wine in new bottle nor a marginal update. Power electronics that initially started its career in processing power revolved primarily around devices like the vacuum tubes, mercury arc rectifiers and, later on, the thyristors. Power electronics has come a long way since then. Technology has improved by leaps and bounds making the power devices more closely to an ideal switch, and the control of the switches are performed in the discrete domain with complex control structures. Thus, power electronics now spans a very wide knowledge base such as power devices, drives, circuit topologies, magnetics, system modeling, control configurations, digital processing, thermal and reliability aspects. If all these aspects are to be treated in a single book, then the topics will, in general, have a superficial bearing. If the topics are to be treated in depth, then it becomes difficult to accommodate all these topics into a single book. Here in lies the challenge and hence this book.

All the topics discussed in this book have been handed down from my teachers and teachers before them by way of exhortation; and to them, I offer my sincere salutations. In the run up to this book, I must say that I have been indeed very lucky to have been tutored by some of the best minds in the field. Every topic has a flavor of the teaching styles of my mentors. From the confluence of the teaching styles of my teachers, I hope that a style will emerge that I may call my own. If the topics are well treated and addressed, then the credit, I must say, should go to my mentors; and if there are mistakes which I presume there will be some, I am solely responsible and will make efforts to correct them in future editions.

The book has been broadly divided into two types of topics: (a) circuit-oriented aspects and (b) system-oriented aspects. The first seven chapters deal with circuit-oriented aspects of power electronic systems, and Chapters 8–15 deal with system-oriented aspects like controls and reliability. Chapter 1 discusses the power semiconductor switches which are the main building blocks of the power electronic systems. The treatment of the semiconductor switches addresses static characteristics, dynamic characteristics, losses and modeling issues. Chapter 2 discusses the drive circuit requirements of BJTs and MOSFETs. Typical drive circuits and their designs are dealt in this chapter. This chapter also introduces the concepts of the series and shunt snubber circuits for power semiconductor devices that behave as power switches.

Chapters 3–6 deal with the power electronic circuit applications. Chapter 3 is on the topic of AC–DC converters or rectifiers. This chapter discusses single- and three-phase rectification topologies along with a detailed discussion on the popular rectifier–capacitor filter circuit addressing its design aspects. The rectifier–LC filter circuit is also discussed. This is followed by a discussion on controlled rectification. Chapter 4 does not fall into the switched-mode category. This chapter addresses the linear regulators wherein the power semiconductor devices are operated in the linear region. Apart from design, the analysis of the linear regulators by progressively including non-idealities is dealt with in a systematic manner.

Chapter 5 discusses the DC–DC converter application. The primary non-isolated topologies of the DC–DC converters are discussed, followed by isolated converters and other special converters. The discussions in this chapter are based on the steady-state analysis that primarily addresses the component design issues. Chapter 6 handles the DC–AC converters or inverters. The various generic topologies are explained followed by a detailed Fourier series analysis of pulse-width modulation strategies for both single- and three-phase inverters. Chapter 7 is a generic topic on magnetic devices that is needed for all applications in AC–DC, DC–DC, DC–AC conversions and isolated drive circuits. This chapter discusses the general principles of magnetism applied to design of inductors, potential transformers and current transformers.

Chapters 8–15 discuss system-oriented aspects of power electronics. Chapter 8 discusses the different modeling methods and their applications to power electronic circuits and systems in order to obtain a dynamic model. Special emphasis is placed on the circuit averaging method, bond graph method and the space vector methods, including detailed discussions with application examples. Chapter 9 is not strictly related to power electronics, but the topics discussed here are essential for designing controllers. It is a cocktail of many topics such as z-transform basics, digital filters, sampling, analog-to-digital conversion methods and performance specification issues.

Chapter 10 develops a formal and systematic approach towards design of controllers for the power electronic systems, wherein the dynamic models are obtained from the concepts of Chapter 8. Both the classical methods of controller design and the state space methods of controller and estimator design are addressed with more emphasis on digital controller and estimator design. Chapter 11 is the extension of the state space controller and estimator design with focus on optimality and robustness issues.

Chapter 12 discusses an important implementation aspect which is discrete computation methods. As most of the control implementations of the power electronic systems are in the digital domain, a detailed discussion on the numeric formats and practical arithmetic algorithms are addressed. This chapter also discusses the implementation of important components like the PI controller and pulse-width modulators within a digital processor. Chapter 13 discusses yet another important and practical aspect of power electronic system which is the thermal aspect. This discusses the heat transfer mechanisms for conducting the heat away from the junction of the power semiconductor devices to the ambient. The selection of heat sinks for the power electronic applications is addressed in this chapter.

Chapters 14 and 15 are devoted to reliability aspects. Chapter 14 discusses in detail the concepts of modeling systems from the reliability point of view. Chapter 15 presents the methods for predicting the reliability of a system. A formal and systematic method to predict the reliability of circuits by part stress co-variate approach is presented by integrating the functional and life aspects of the circuit specifications at the design stage itself. A MATLAB based toolbox called reliability for electronic circuits (REC) is included in the accompanying CD. This toolbox is developed based on the concepts presented in Chapter 15.

As is evident from the topics and the chapters, the vast area of power electronics cannot be handled in a single semester course of the engineering curriculum. In order to handle the topics for semester duration, the chapters may be re-organized in the following manner that is only suggestive and also not exhaustive.

- Parts of Chapters 1, 3, 4 and 7 are a possible combination for a course on linear power supplies.
- Parts of Chapters 1–3, 5 and 7 are a possible combination for a course on switched-mode DC–DC power supplies.
- Chapters 5 and 7 can be suitable for a course on DC–DC converters.
- Parts of Chapters 1–3, 6 and 7 are a possible combination for a course on inverters.
- Chapters 6 and 7 can be suitable for a course on inverters along with pulse-width modulation strategies.

- Chapters 5, 8–10 can be useful for an advanced-level course on dynamics of DC–DC converters.
- Chapters 8–11 by themselves can be suitable for an advanced-level course on control of power electronic systems.
- Chapters 5, 7, 12 and 13 can be a possible combination for a course on implementation aspects of DC–DC converters.
- Chapters 6, 7, 12 and 13 can be a possible combination for a course on implementation aspects of inverters.
- Chapters 14 and 15 by themselves can be useful for a course on reliability of power electronic systems.
- Parts of Chapters 5, 7, 13–15 can be suitable for a course on reliability design for DC–DC converters.
- Parts of Chapters 6, 7, 13–15 can be suitable for a course on reliability design for inverters.

I must acknowledge that though it appears that I am the sole author of this book, the material in the book has evolved over many years with constant interaction with my teachers, colleagues and students who have directly and indirectly contributed to the knowledge base of the book; my sincere salutations to all of them.

For the past year and a half, I have not given sufficient time to my family and the matters of the home due to the long and late hours spent in writing this book. My wife and son have been very patient and supportive in this aspect, awaiting the time for the completion of the book. I sincerely acknowledge my gratitude for their continuous support.

L. Umanand

Contents

Foreword	v
Preface	vii

1	Power Semiconductor Switches	1
1.1	Ideal Switch	2
1.2	Diodes	2
	Static Characteristics	6
	Dynamic Characteristics	7
	Diode Classifications	8
	Diode Parameters	8
1.3	Surge Current	9
1.4	Thermal Viewpoint	9
	Diode Model	10
1.5	Bipolar Power Transistors	11
	Static Characteristics	13
	Dynamic Characteristics	14
	Calculation of Power Dissipation in a Transistor	17
1.6	Safe Operating Area (SOAR)	19
1.7	Paralleling of Transistors	20
1.8	Darlington Connection	21
1.9	Power MOSFETs	22
	Static Characteristics	25
	Dynamic Characteristics	25
	Paralleling of MOSFETS	29
	Handling Precautions for MOSFET	29
	Remarks on BJT and MOSFET	29
	Power Dissipated in a Power MOSFET	30
	Model for Power MOSFET	30
1.10	Insulated Gate Bipolar Transistor (IGBT)	30
	Static Characteristics	34
	Dynamic Characteristics	35
	Remarks on IGBT	37
1.11	Thyristor Family	37
	Silicon-Controlled Rectifier (SCR)	38
	DIACs and TRIACs	42
	Gate Turn-OFF Thyristor (GTO)	44
	MOS-Controlled Thyristor (MCT)	46
	Concluding Remarks	46
	Laboratory Exercises	47
	Fill in the Blanks	51

	Descriptive Questions	52
	Problems	53
	Answers	53
2	Drive Circuits	55
2.1	Transistor Drive Circuits	55
	Turn-ON Behavior	56
	Turn-OFF Behavior	56
	Characteristics and Classification of the Drive Circuits	58
	BJT Drive Circuit-1	59
	BJT Drive Circuit-2	59
	BJT Drive Circuit-3	60
	BJT Drive Circuit-4	61
	BJT Drive Circuit-5	61
	BJT Drive Circuit-6	62
	BJT Drive Circuit-7	64
	BJT Drive Circuit-8	65
	BJT Drive Circuit-9	65
	BJT Drive Circuit-10	66
	BJT Drive Circuit-11	67
	BJT Drive Circuit-12	67
	BJT Drive Circuit-13	69
2.2	MOSFET Drive Circuits	70
	MOSFET Drive Circuit-1	71
	MOSFET Drive Circuit-2	72
	MOSFET Drive Circuit-3	73
	MOSFET Drive Circuit-4	73
	MOSFET Drive Circuit-5	74
	MOSFET Drive Circuit-6	75
	MOSFET Drive Circuit-7	75
	MOSFET Drive Circuit-8	76
	MOSFET Drive Circuit-9	77
	MOSFET Drive Circuit-10	78
	MOSFET Drive Circuit-11	79
2.3	Snubber Circuits	80
	Turn-OFF Snubber or Shunt Snubber	80
	Turn-ON Snubber or Series Snubber	83
	Concluding Remarks	85
	Laboratory Exercises	86
	Fill in the Blanks	91
	Descriptive Questions	92
	Problems	93
	Answers	94

3	Rectifiers	95
3.1	Uncontrolled Rectifiers	96
3.2	Rectifier Circuits	96
	Single-Phase Circuits	96
	Three-Phase Circuits	103
3.3	Capacitor Input Filter	109
	Design of Capacitor Input Filter Rectifier	110
	Turn-ON Currents and Surge Limiting	117
3.4	Power Factor	121
3.5	Rectifier–LC Filter	124
	Output Ripple	127
	Turn-ON Current	128
	Design Summary	128
3.6	Controlled Rectifiers	128
	Single-Phase Power Circuits	129
	Three-Phase-Controlled Rectifier Circuits	134
	Concluding Remarks	142
	Laboratory Exercises	142
	Fill in the Blanks	145
	Descriptive Questions	146
	Problems	147
	Answers	148
4	DC–DC Linear Regulators	149
4.1	Generic Linear Regulator	149
	Operating Principle	150
4.2	Linear Regulator Topologies	151
	Shunt Regulator	151
	Modified Shunt Regulator	154
	Series Regulator	154
	Negative Voltage Regulator	156
	IC Linear Regulators	158
	Practical Tips	160
	Four-Wire Connection	162
	Protection	163
	Current Regulation	165
4.3	Parameters of the Linear Regulator	166
4.4	Analysis of a Series Regulator	166
	First-Level Modeling	166
	Effect of Current through R_b	170
	Effect of Current through R_l	171
	Regulator Parameters	172
	Range of Load Resistance R_L	173

	Second-Level Modeling	174
	Influence of Temperature	179
4.5	Analysis of a Current Regulator	180
	Line Regulation	182
	Range of R_L	182
	Second-Level Modeling	183
	Concluding Remarks	184
	Laboratory Exercises	184
	Fill in the Blanks	188
	Descriptive Questions	188
	Problems	189
	Answers	192
5	DC–DC Switched-Mode Converters	193
5.1	Single Pole Double Throw Switch	193
5.2	The Chopper	194
5.3	DC Steady-State Principles	200
5.4	The Three Basic Converters	201
	Step-Down or Buck Converter	201
	Step-Up or Boost Converter	206
	Step-Up/Down or Buck–Boost Converter	213
5.5	Selection of Power Switches	219
	Device Selection	219
	Electrical Stress Rating	221
	Thermal Stress Rating	221
5.6	Continuous and Discontinuous Conduction Modes	221
5.7	Isolated Converters	225
	Forward Converter	226
	Semiconductor Switches	228
	Waveforms	229
	Governing Equations	230
	Forward Converter with Demagnetizing Winding	233
	Dual Switch Forward Converter	236
	Push–Pull Converter	238
	Half-Bridge Converter	243
	Full-Bridge Converter	248
	Flyback Converter	253
5.8	Practical Issues	258
5.9	Other Topologies	258
	Cuk Converter	259
	High-Boost Converter	263
	Soft Switching in Converters	265
	Concluding Remarks	275
	Laboratory Exercises	276
	Fill in the Blanks	281

	Descriptive Questions	283
	Problems	286
	Answers	288
6	DC–AC Switched-Mode Converters	289
6.1	Inverter Topologies	290
	Single-Phase Topologies	290
	Three-Phase Topologies	296
6.2	Self-Driven Inverters	297
	Saturable-Core Inverter	298
	Saturable Base Drive Inverter	302
6.3	Driven Inverter	304
	Push–Pull Configuration	304
	Half-Bridge Configuration	306
	Full-Bridge Configuration	308
6.4	Quasi-Square Wave Inverter	309
6.5	Three-Phase Inverter Topologies	312
	Six-Step Inverter	314
	Current-Controlled Inverter	315
6.6	Pulse-Width Modulation	318
	Quasi-Square Modulation	321
	End Pulse Modulation	324
	Center Pulse Modulation	327
	Sinusoidal Pulse-Width Modulation	329
	PWM through Phase Modulation	337
	Space-Vector Modulation	339
6.7	Pulse–Width Modulation with Harmonic Elimination	350
	Harmonic Elimination with Multiple Notches	351
	Determination of Notch Angles	352
	Staircase PWM	355
	Summary of Staircase PWM Design	357
	Concluding Remarks	358
	Laboratory Exercises	359
	Fill in the Blanks	363
	Descriptive Questions	365
	Problems	367
	Answers	368
7	Design of Magnetics	369
7.1	Magnetic Concepts	369
	Dissipative Component	371
	Flux and Flux Density	372
	Volt-Second Balance	374

7.2	Inductor	375
	Inductor Value	375
	Energy Storage	375
	Area Product	377
	Window Area	378
	Cross-Section Area	378
	Design of Inductors	379
	Multiple-Winding Inductors	382
7.3	Potential Transformer	383
	Operating Principle	384
	Turns Ratio	385
	Leakage Flux Linkage	386
	Equivalent Circuit	388
	Area Product	391
	Transformer Design	394
7.4	Current Transformer	398
	Design of Current Transformers	401
7.5	General Notes on Magnetics	403
	Window Utilization Factor	403
	Core Shapes	404
	Polarity and Dot Convention	406
	Magnetic Losses	407
	Skin Effect	412
	Proximity Effect	413
	Concluding Remarks	415
	Laboratory Exercises	416
	Fill in the Blanks	418
	Descriptive Questions	419
	Problems	420
	Answers	421
8	Modeling of Systems	423
8.1	Input–Output Relations	425
8.2	Differential Equations and Linearization	426
8.3	State Space Representation	428
8.4	Transfer Function Representation	431
	Modeling an Armature-Controlled DC Motor	434
	Poles and Zeros	436
8.5	Block Diagrams	436
	Mason’s Rule	440
8.6	Lagrange Method	442
8.7	Circuit Averaging	447
8.8	Bond Graphs	453
	Standard Elements	454
	One-Ports	454

	Two-Ports	455
	Multi-Ports (Junctions)	457
	Rules for the Selection of Causality	457
	Steps in Obtaining the System Model	457
	Bond Graph Construction	458
	Causality Assignment	462
	State Equation Extraction	463
	Modeling Switched Power Systems	467
8.9	Space-Vector Modeling	473
	Space Vectors	475
	Representation of Space Vectors in Orthogonal Co-ordinates	476
	Space-Vector Transformations	476
	Modeling of Induction Motor	479
	State Space Representation of the d–q Model of the Induction Motor	483
	Concluding Remarks	491
	Tutorial Exercises	492
	Fill in the Blanks	494
	Descriptive Questions	496
	Problems	497
	Answers	498
9	Control System Essentials	499
9.1	Representation of System in Digital Domain	501
9.2	The z-Transform	502
	Physical Significance of z^{-1}	506
9.3	Digital Filter	508
	Few Common Digital Filters	512
9.4	Mapping between s-Plane and z-Plane	515
	$s = 0 + j0 \rightarrow z = 1$	515
	$s = 0 + j\pi/T \rightarrow z = -1$	516
	$s = 0 - j\pi/T \rightarrow z = -1$	516
	$s = -\sigma + j0 \rightarrow z = \text{Real Axis between 0 and 1}$	517
	$s = \sigma + j0 \rightarrow z = \text{Real Axis between 1 and Infinity}$	517
9.5	Effect of Sampling	518
9.6	Continuous to Discrete Domain Conversion	521
	Pole–Zero Map	522
	Numerical Integration Methods	523
	Hold Equivalence	530
	Discretizing the State Equation	533
9.7	Control System Basics	535
	Performance Specifications	535
	Stability	535
	Steady-State Accuracy	536
	Speed of Response	538
	Measures for Performance Parameters	538

9.8	Control Principles	540
9.9	State Space Method	542
	Concluding Remarks	543
	Tutorial Exercises	544
	Fill in the Blanks	545
	Descriptive Questions	546
	Problems	547
	Answers	548
10	Digital Controller Design	549
10.1	Controller Design Techniques	550
10.2	Bode Diagram Method	551
	Bode Diagrams	552
	Controller Design with Bode Diagram	556
10.3	PID Controller	562
	PID Controller Design	567
	Integrator Windup in PID Controller	568
	Practical PID Controllers	570
10.4	Root Locus Method	573
	Root Locus Plot	574
	Controller Design	575
10.5	State Space Method	582
	Controllability	582
	Observability	583
10.6	Full-State Feedback	584
	Regulator	585
	Tracker	585
10.7	Regulator Design by Pole Placement	587
10.8	Estimator Design	591
	Estimator Error Dynamics	592
	Separation Principle	593
	Current Estimator	594
	Reduced Order Estimator	596
10.9	Tracker: Controller Design	599
10.10	Controlling Voltage	600
	Operation	605
10.11	Controlling Current	607
	Current-Controlled Converter	607
	Unity Power Factor Converter	610
	Front-End Converter	613
10.12	Control of Induction Motor	619
	Scalar Control	619
	Vector Control	622
10.13	Output Feedback	631

10.14	Induction Motor Control with Output Feedback	633
	Plant Dynamics	636
	Anti-Alias Filter Dynamics and Augmentation	636
	Controller Dynamics and Augmentation	638
	Concluding Remarks	641
	Tutorial Exercises	642
	Fill in the Blanks	643
	Descriptive Questions	644
	Problems	646
	Answers	647
11	Optimal and Robust Controller Design	649
11.1	Least Squares Principle	650
11.2	Quadratic Forms	651
11.3	Minimum Energy Principle	653
11.4	Least Squares Solution	655
11.5	Weighted Least Squares	658
11.6	Recursive Least Squares	659
	LMS Algorithm	660
	Kalman Filter	664
11.7	Optimal Control: Linear Quadratic	668
	Linear Quadratic Regulator	670
	Linear Quadratic Gaussian	672
	Linear Quadratic Tracker: Output Feedback	672
11.8	Induction Motor Example	674
11.9	Robust Controller Design	679
	Frequency Domain Performance Specifications	681
	Modeling Uncertainties	681
	Controller Design Steps	682
	Speed Controller Synthesis	685
	Concluding Remarks	688
	Tutorial Exercises	688
	Fill in the Blanks	691
	Descriptive Questions	692
	Problems	693
	Answers	694
12	Discrete Computation Essentials	695
12.1	Numeric Formats	695
	Fixed-Point Numeric Format	696
	Floating-Point Numeric Format	699
	General Notes on Number System	702

12.2	Tracking the Base Point in the Fixed-Point System	703
	Addition of Numbers	703
	Subtraction of Numbers	703
	Multiplication of Numbers	703
12.3	Normalization and Scaling	704
	Multiplication Algorithm	707
12.4	Arithmetic Algorithms	709
	Reciprocal	709
	Square Root	711
	Reciprocal of Square Root	713
	Hypotenuse	713
	Polynomial	714
	Sine and Cosine	714
	Exponential	715
	Logarithm	716
	Implementation Examples	717
	PI Controller	717
	Sine and Cosine	719
	Pulse-Width Modulation (PWM)	720
	Space Vector PWM	724
	Over-Modulation	727
	Concluding Remarks	729
	Tutorial Exercises	730
	Fill in the Blanks	732
	Descriptive Questions	732
	Problems	733
	Answers	734
13	Thermal Aspects	735
13.1	Modeling Heat Transfer	736
13.2	Heat Transfer by Conduction	738
13.3	Heat Transfer by Convection	742
	Determination of Nusselt Number	743
	Free Convection	744
	Forced Convection	747
13.4	Heat Transfer by Radiation	753
	Reflectance, Absorptance and Transmittance	755
13.5	Heat Transfer by Mass Transport	756
13.6	General Notes on Heat Transfer	758
	Mixed Heat Transfer Mechanisms	758
	Thermal Resistance of Varying Cross-Sections	759
	Mounting of Power Devices on Heat Sink	760
	Transient Model for Heat Transfer	761
	Best Method of Heat Transfer	762

	Concluding Remarks	763
	Tutorial Exercises	763
	Fill in the Blanks	765
	Descriptive Questions	766
	Problems	767
	Answers	768
14	Reliability Modeling	769
14.1	Reliability Concepts	771
	Failure Rate	771
	Failure-Time Probability Density Function	773
	Mean Time Between Failure (MTBF), Mean Time to Failure (MTTF)	776
14.2	Modeling for Reliability	777
	Series System	777
	Parallel System	778
	Mesh System	786
14.3	Redundant Systems	788
	Full-On Redundancy: Multiple-Survivor Sub-System	789
	Switching in Redundant Systems	791
	Mean Time to Failure	795
	Standby Redundancy	797
	Standby Redundancy with Imperfect Switching	799
14.4	Redundant Systems with Non-Identical Components	799
14.5	Design for Reliability	804
14.6	Reliability Apportioning/Allocation	813
	Method 1: Equal Apportionment Technique	814
	Method 2: AGREE Apportionment Technique	814
	Method 3: ARINC Apportionment Technique	815
14.7	Inter-Connection and Wiring	816
14.8	General Notes on Designing for Reliability	822
	Connectors	822
14.9	Part Derating	828
	Concluding Remarks	828
	Tutorial Exercises	829
	Fill in the Blanks	830
	Descriptive Questions	831
	Problems	832
	Answers	833
15	Reliability Prediction	835
15.1	Similar Equipment Technique	836
15.2	Similar Complexity Technique	836
15.3	Prediction by Function Technique	836

15.4	Parts Count Technique	837
	Part Count with Non-Series Model	839
15.5	Part Stress Technique	839
15.6	Environments	842
15.7	Reliability Prediction Process	842
15.8	Component Naming Convention	843
15.9	Component Object Notation	844
15.10	Example Reliability Prediction Process	849
	Steps to Design a Circuit	849
15.11	Physics of Failure (PoF)	852
	Concluding Remarks	854
	Tutorial Exercises	854
	Fill in the Blanks	855
	Descriptive Questions	855
	Problems	856
	Answers	857
	 Appendix I	 859
	Appendix II	861
	Appendix III	863
	Appendix IV	865
	Appendix V	866
	Appendix VI	867
	 Bibliography	 909
	Index	911

Power Semiconductor Switches

Learning Objectives

After reading this chapter, you will be able to:

- understand the requirements of an ideal switch and the characteristics of important power semiconductor switches.
 - estimate the conduction and switching power losses in various power semiconductor switches.
 - model and simulate the power semiconductor switches.
-

Power electronic applications deal with the flow of power. However, majority of the applications in power electronics are based on switching the power flow in order to improve efficiency. As a consequence, one has to use power semiconductor devices that act as power switches. This chapter discusses various power semiconductor devices with an emphasis on using them as power electronic switches. There are many semiconductor devices that can be used as power switches. Few of the common power semiconductor switches are diodes, bipolar junction transistors (BJTs), metal oxide semiconductor field effect transistor (MOSFETs), insulated gate bipolar transistors (IGBTs), thyristors, gate turn-OFF thyristors (GTOs) and metal oxide semiconductor controlled thyristor switches (MCT). Power MOSFETs and IGBTs have a large role to play in modern power control equipments and hence their study is emphasized.

The power semiconductor switches may be studied from various viewpoints:

1. physics viewpoint;
2. circuit viewpoint;
3. protection viewpoint;
4. drive viewpoint;
5. modeling viewpoint;
6. packaging viewpoint.

The **physics** viewpoint explains the operation and the functional features of the device. The **circuit** viewpoint deals with the static and dynamic characteristics. For reliable operation of the power switches, one must ensure that the electrical and thermal stresses within the device are well below the stated ratings of the device. These switches handle large currents and dissipate a considerable amount of heat; consequently, the thermal aspects need detailed attention to ensure that the switch operates within the permissible junction temperatures. The **protection** viewpoint focuses on the electrical and thermal stresses within the

device. In this respect, the safe operating limits for the operation of the power semiconductor device are addressed. The **drive** viewpoint emphasizes and clarifies the switching behavior of the power devices that enables one to synthesize reliable and meaningful drive circuits. The **packaging** viewpoint focuses on the device mounting strategies, removal of heat through heat sinks, forced cooling devices and connection issues.

Modeling switches for purposes of simulation are also important aspects that need to be addressed. The mathematical representation of the switches will aid in simulating the power topologies so that the circuit behavior and the waveforms at various parts in the circuit may be determined before the actual hardware implementation. In fact, it is essential to complement the hardware bread-boarding sessions with electronic bread-boarding sessions not only to enhance the understanding of the circuit and its operation, but also to reduce the design cycle time.

1.1 Ideal Switch

This chapter focuses on semiconductor devices that are operated in such a manner that they behave as switches. These semiconductor switches are supposed to emulate the operation of an ideal single pole single throw (SPST) switch. However, it will become clear from the discussions in the sections to follow that none of the semiconductor switches have all the characteristics of an ideal SPST switch. Before discussing the features of the semiconductor switches, it is necessary to have a reference list of features that an ideal SPST switch has in order to aid in a better understanding of the practical semiconductor switch. An ideal SPST switch will have the following features:

1. ON resistance = 0 (or zero forward voltage drop).
2. OFF resistance = infinity (or zero reverse current).
3. When ON-conducts infinite current in both the forward and reverse directions.
4. When OFF-withstands infinite forward and reverse voltages.
5. It can switch instantaneously from OFF to ON and from ON- to OFF-states.
6. Power dissipated in the switch is zero, that is, both the conduction and the switch transition losses are zero.
7. ON-to-OFF and OFF-to-ON transitions of the switch are fully controllable.
8. It requires no power to drive or control the switch.

While none of the power semiconductor switches have these ideal characteristics, efforts are continuously made to improve the performances of the semiconductor switches such that they may tend towards ideal behavior. There are many different types of power semiconductor devices available commercially. However, in this chapter, only a few popular generic power semiconductor switch types will be discussed. The discussion will initially try to provide some basic insights into the physics of operation of the semiconductor junction taking diode as an example. Subsequently the circuit viewpoint is discussed for the diode as well as few other device types like BJTs, MOSFETs, IGBTs and thyristors to aid in the selection of the devices as power switches for power electronic applications.

1.2 Diodes

Diodes are devices that have two terminals, namely, (a) anode and (b) cathode. When the anode is more positive than the cathode, the diode is said to be **forward-biased** and allows a flow of current from the anode to the cathode. The forward-biased state of the diode is called the ON-state or the conducting state

of the diode. On the other hand, when the cathode is more positive than the anode, the diode is said to be **reverse-biased** and does not permit any flow of current. The reverse-biased state of the diode is called the OFF-state or the blocking state of the diode. The switching action of the diode is solely dependent on the anode-to-cathode potential and this is determined by the external circuit. As there is no control on the switching state, a diode is termed as an **uncontrolled** switch.

In a metal (e.g. copper), the valence electrons are completely free and roam from atom to atom. In a semiconductor (e.g. silicon), a covalent bond is imposed between the valence electrons of adjacent atoms. As a result the motion of valence electrons is coordinated with the motion of valence electrons of an adjacent atom. Thus, in a pure semiconductor there are very few free electrons. Evidently, the conduction property of a metal and a semiconductor is very different. If a fraction of a volt is applied across a semiconductor, a small amount of current will flow whereas if applied across a metal, a large current will result.

The conductivity of pure semiconductors can be changed by adding impurities. This process of adding impurities into a pure semiconductor is called **doping**. By adding some impurities in a pure semiconductor, its conductivity can be increased and electron (or hole) flow can be easily controlled, for example, pure silicon plus a pentavalent impurity (e.g. arsenic) increases the mobile free electron charges and is called ***n*-material**, whereas pure silicon plus a trivalent impurity (e.g. boron) increases the mobile hole charges and is called as ***p*-material**.

A *p-n* junction diode can be formed by growing a single crystal of semiconductor material and doping with the above impurities in a controlled manner. Figure 1.1 shows a schematic diagram of a *p-n* junction. The *p*-side has a high density of holes (i.e. electron vacancies) and the *n*-side has a high density of electrons. These electrons will diffuse from the *n*- to the *p*-side and the holes will diffuse from the *p*- to the *n*-side. This will create a space charge layer on either side of the junction called the depletion region wherein the holes and the electrons combine to form immobile charges as shown in Figure 1.1. This gives rise to an electric field and a potential barrier. The electric field creates a drift current that acts in such a way as to oppose the diffusion current. Equilibrium is reached when the diffusion and the drift currents balance each other.

The depletion region at the junction has only immobile charges. The value of the potential barrier in volts depends on the charge carriers and is governed by the Boltzmann's relation given in Eq. (1.1). The

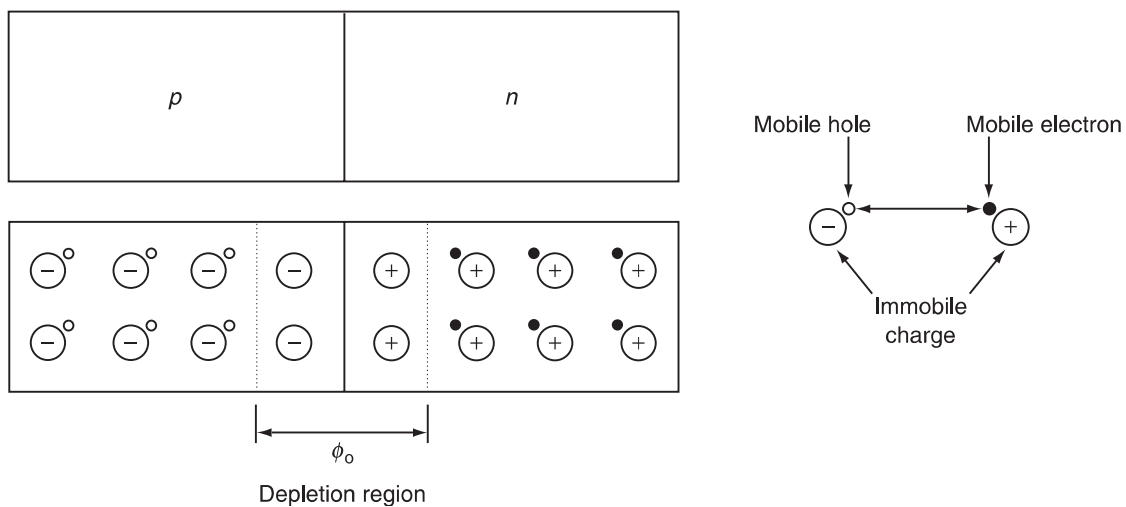


Figure 1.1 | Schematic diagram of *p-n* junction.

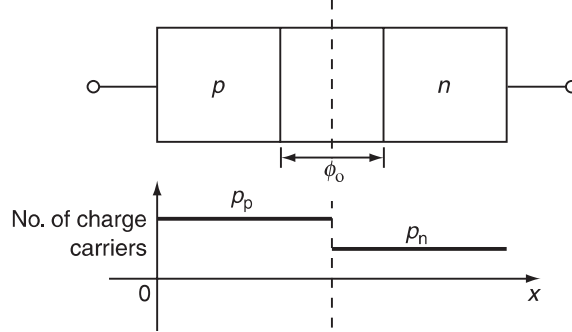


Figure 1.2 | Charge carrier levels in the p - and n -regions in the absence of external bias.

charge carrier levels at equilibrium for the p -region and the n -region in the absence of any external bias voltage are depicted in Figure 1.2. The number of holes in the p -region is denoted by p_p and the number of holes in the n -region is denoted by p_n :

$$p_n = p_p \times e^{-\phi_0/(KT/q)} = p_p e^{-\phi_0/V_T} \quad (1.1)$$

where $V_T = KT/q$ and K is the Boltzmann constant in Joules per degree Kelvin; T is the junction temperature in degree Kelvin; q is the electron charge in coulomb; p_n is the number of holes in the n materials; p_p is the number of holes in the p material; ϕ_0 denotes the barrier potential.

Note that Eq. (1.1) is an exponential relationship. V_T is of the order of few tenths of volt. With this basic relationship one can try to understand the operation of a p - n junction that is the building block of almost all semiconductor devices.

On applying a forward bias (positive to p and negative to the n) to the p - n junction as indicated in Figure 1.3(a), the barrier potential across the p - n junction reduces from the equilibrium value of ϕ_0 to $(\phi_0 - \phi)$ and the resulting equilibrium hole density in the n -material is changed, which is given by Boltzmann law as

$$p_n^* = p_p \times e^{-(\phi_0 - \phi)/(KT/q)} = p_n e^{\phi/V_T} \quad (1.2)$$

Equation (1.2) is a fundamental relationship in junction theory. This shows that a small forward bias increases the minority carrier (holes) concentration in the n -region exponentially. These excess holes come from the p -side and are in turn replenished from the external source. As the holes cross over from the p -region to the n -region, swift recombination results. The charge density decreases with distance, x from the junction as depicted in Figure 1.3(a). For every electron in the n -material that combines with a hole, there is another electron entering the n -region from the negative side of the external voltage source. It is important to understand that this change of charge density with distance in the semiconductor involves a transport of charge and thereby constitutes a current flow. It takes place predominantly by diffusion.

The hole-current through diffusion is proportional to dp_n^*/dx . Similarly diffusion current due to the electron density difference also exists. The diffusion current in the p - n junction is given as

$$I = I_0 (e^{qV/KT} - 1) \quad (1.3)$$

where I is the diode current; I_0 the saturation current; q the electronic charge in coulombs; K the Boltzmann's constant; T the junction temperature in degree Kelvin.

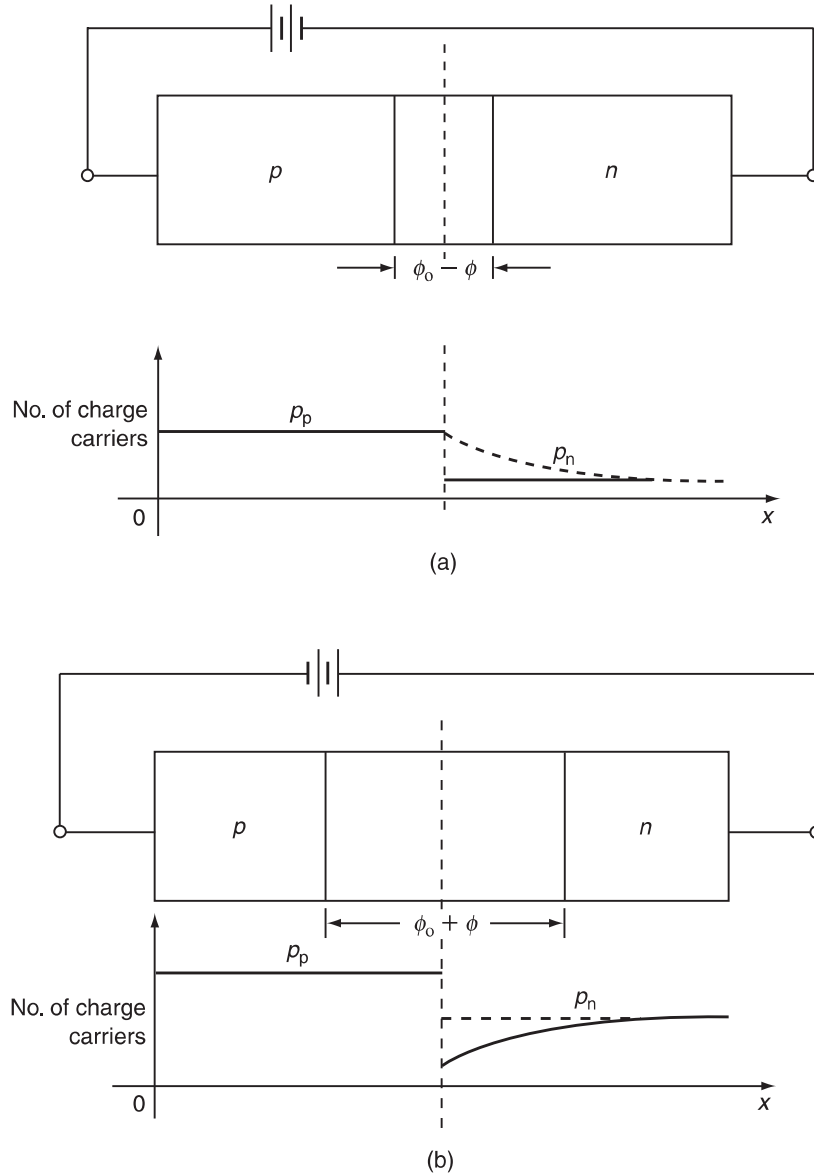


Figure 1.3 | Charge densities near a biased p - n junction.

In the conventional conductor, the current and the voltage are related by the well-known Ohm's law. However, it is evident from the expression given in Eq. 1.3, that in the case of a p - n junction, the current is exponentially related to the voltage as a consequence of the Boltzmann relation. On applying a reverse bias (positive to n and negative to p) to the p - n junction as indicated in Figure 1.3(b), the barrier potential across the p - n junction increases to $(\phi_0 + \phi)$ and the resulting equilibrium hole density in the n -material is given by the Boltzmann law as

$$p'_n = p_p e^{-(\phi_0 + \phi)/(KT/q)} = p_n e^{-\phi/V_T} \quad (1.4)$$

Under reverse-biased condition, p'_n is less than the equilibrium density of holes in the n -region. This implies that the holes diffuse from the n -region into p -region. Thus, an extremely small current called the reverse saturation current results and is denoted as I_0 . A very large negative voltage disrupts all covalent bonds and results in a large current. Now it behaves like a metallic conductor and the diode is said to have attained reverse breakdown.

Static Characteristics

The symbol of a diode is shown in Figure 1.4(a). Lead A that is connected to the p -region of the p - n junction is called the anode and lead K that is connected to the n -region of the p - n junction is called the cathode. The v - i characteristics of an ideal diode are depicted in Figure 1.4(b). From Figure 1.4(b), it can be observed that the voltage across the diode when forward-biased is zero. In forward-biased condition it allows flow of currents from the anode to the cathode. When the diode is reverse-biased, then the diode blocks any flow of current. However, the diode can withstand the applied reverse voltage as shown in Figure 1.4(b).

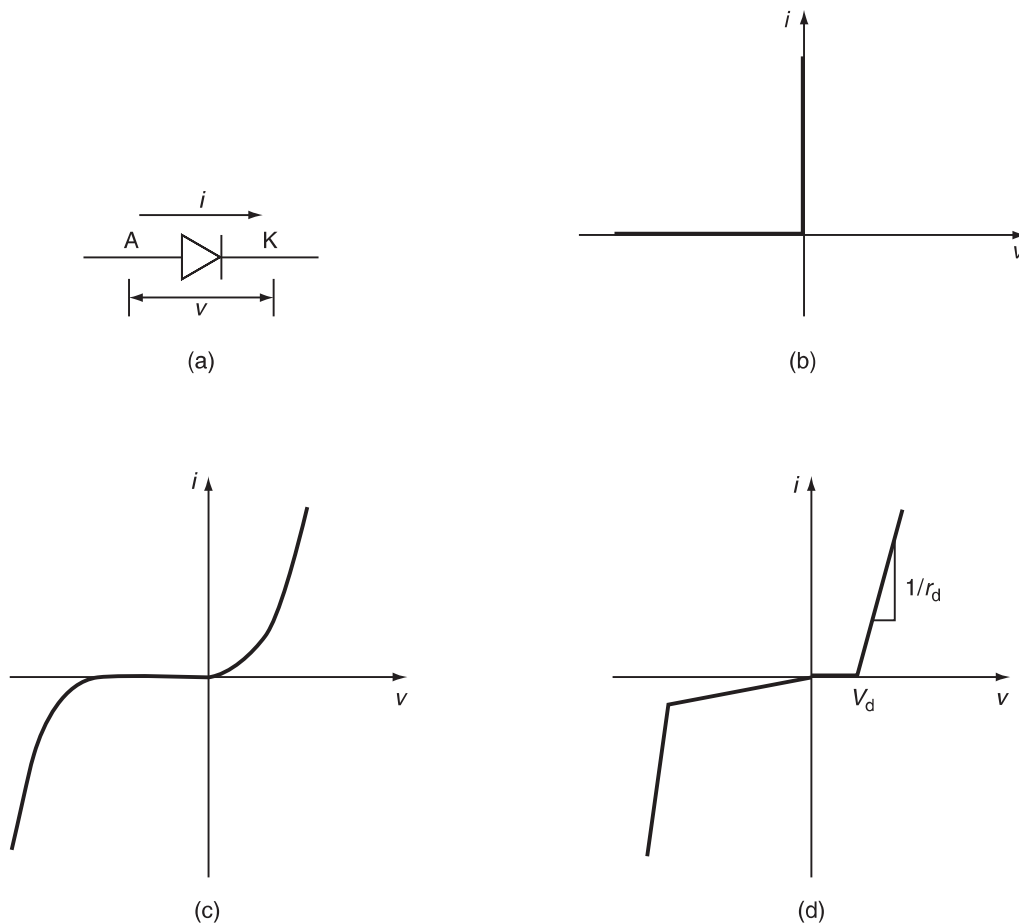


Figure 1.4 (a) Symbol of diode; (b) ideal v - i characteristics; (c) actual characteristic; (d) piece-wise linear characteristic.

The practical v - i characteristic of the diode is shown in Figure 1.4(c). It shows that during the forward-bias condition, the diode has a finite resistance called the forward dynamic resistance (r_d). For quick engineering calculations, it is normal to simplify the characteristics by the piece-wise linear characteristic, as shown in Figure 1.4(d).

Dynamic Characteristics

Turn-OFF of Diode

A forward-biased diode has a charge distribution as shown in Figure 1.3. If a reverse bias is now applied to the diode, the charges will have to re-distribute from p_n^* to p_n' . This takes a certain amount of time. The waveform for the current and voltage are shown in Figure 1.5. The test circuit used to obtain the dynamic characteristics is as shown in Figure 1.5(a).

The excess charge stored in the diffusion region has to be removed before the junction can be reverse-biased. As long as there are excess charge carriers in the diffusion region (also called the space-charge region), the junction will be in forward-biased state. The diode voltage will not change from its ON-state value except for the small decrease due to Ohmic drop caused by the reverse current. After the current goes negative and the excess charges are removed at time t_2 , the junction becomes reverse-biased and quickly acquires the applied negative voltage value. At time t_2 , the junction charge distribution would have reached the equilibrium charge distribution of a non-biased junction like that shown in Figure 1.2. From t_2 to t_3

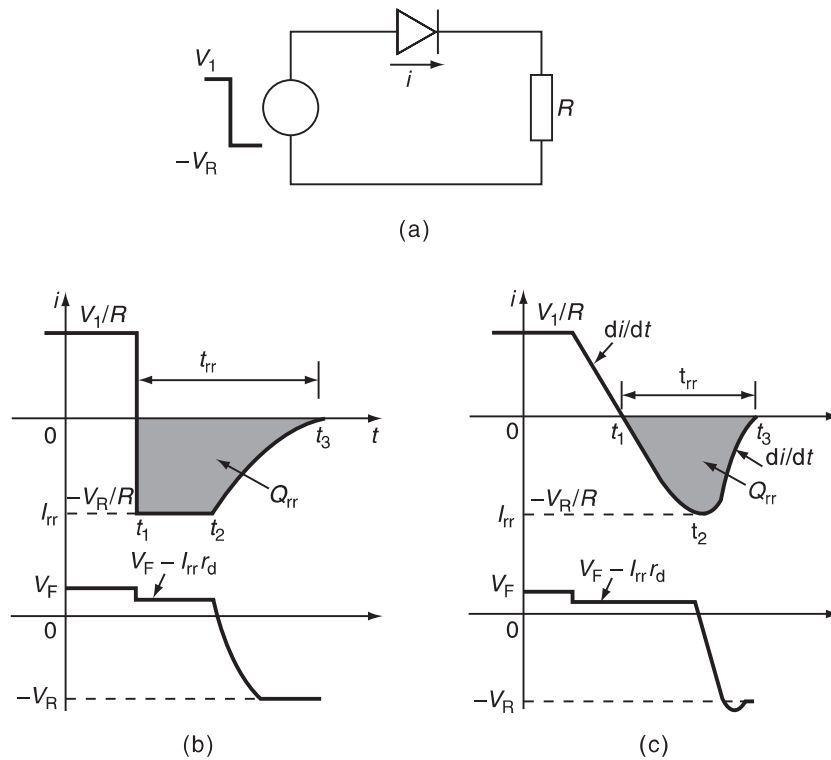


Figure 1.5 | Diode turn-OFF: (a) Test circuit for turn-OFF; (b) idealized dynamic characteristic during turn-OFF; (c) practical dynamic characteristic during turn-OFF.

the charge distribution tends towards that of the reverse-biased junction like that shown in Figure 1.3(b). The diode now starts to withstand the applied reverse voltage and the diode current falls quickly to zero. In the case of a practical diode, owing to parasitic inductance, the current in the diode does not immediately reverse like that of the ideal recovery shown in Figure 1.5(b). It reduces at a certain rate decided by the lead inductance as indicated in Figure 1.5(c). The time interval t_{rr} is called the reverse recovery time and is an important parameter in switching applications. The interval between t_1 and t_2 is sometimes called the storage time.

Turn-ON of Diode

If the diode is under reverse bias and it has to be forward-biased, it requires certain time, known as turn-ON time or forward recovery time before all carriers in the whole junction can contribute to the current flow. Here, the charges will have to re-distribute from p'_n to p_n^* . In the case of turn-OFF, the charges have to be removed or re-combined whereas in the case of turn-ON the junction has to acquire charges. This is generally a faster process than removal of the charges. Therefore, turn-ON times will be much faster than turn-OFF times. In practical situations, it turns out that forward recovery time does not constitute a serious problem.

Diode Classifications

The diodes are generally classified based on the turn-OFF times. The reverse recovery time (t_{rr}) is a measure of the speed at which the diode can switch (turn-OFF) and therefore gives an indication on the external switching frequencies that may be used. Based on this, the diodes are classified as follows:

1. Rectifier diodes (t_{rr} in microsecond range).
2. Fast recovery diodes (t_{rr} in 200–500 ns).
3. Ultra fast recovery diodes ($t_{rr} \sim 30$ –200 ns).
4. Schottky diode (metal semiconductor junction $-t_{rr} < 30$ ns).

In high-frequency switching circuits, fast recovery diodes (types 2–4) should be used depending on the frequency range of interest and voltage rating. Type 1 is used for low-frequency applications like mains rectification, as in front-end rectifier for the capacitor input filter.

Diode Parameters

There are various parameters in the datasheet of a diode that need to be understood before selecting a diode for a particular type of application. These parameters must be calculated *a priori* before selecting a diode for the specific application. The important parameters in the datasheet of a diode are

1. Average current, I_{Fav}
2. Root mean square (RMS) current, I_{Frms}
3. Peak current, I_F
4. Surge current, I_{FSM}
5. Breakdown or reverse voltage, V_{RRM}
6. Forward drop, V_F
7. Dynamic resistance, r_d
8. Reverse recovery time, t_{rr}
9. I^2t rating

For sinusoidal currents, I_{Fav} , I_{Frms} and I_{F} are related and the relationships are well known. It is not so when current in the circuit is non-sinusoidal. In such cases, depending on the current and voltage waveforms of the diodes for the specific circuitry, the corresponding average, rms and peak values must be calculated from the fundamentals.

In high-power circuits, it is important to make sure that the chosen diode ratings are higher than the respective values of currents that will flow in the circuit to ensure that the diode can reliably operate. In detailed datasheets, some of the parameters may be given in the form of nomo-graphs. One may choose the diode based on the parameter nomo-graphs also.

1.3 Surge Current

The rectifier has the ability to sustain overload currents for a brief period of time without damaging the device. Two parameters are specified in datasheets: I_{FSM} and I^2t rating. I_{FSM} is the maximum peak half-cycle non-repetitive current rating. In applications that use the capacitor input rectifier filter, the problem of surge current is severe. When the capacitor is fully discharged, the surge current will be enormous at the time when the rectifier filter is switched ON and the input voltage waveform is at its peak. This problem is most severe in the case of line-operated switched-mode power supplies when the input or the mains voltage is at its worst case maximum value of around 270 V rms. If the expected surge current is higher than what the device can sustain, one has to limit the current using some limiting device such as a series resistor. The amp-squared second (I^2t) rating is normally associated with protective fuse co-ordination and the rating is usually given for 10 ms. For proper functioning, the fuse I^2t rating has to be less than the diode I^2t rating.

1.4 Thermal Viewpoint

An ideal diode does not have any power dissipation. However, a practical diode will have a power loss. It is essential to quantify the amount of power loss in the diodes so that one can select a suitable heat sink in addition to budget the power loss in the circuit so that the efficiency can be estimated. There are different components of power loss in a diode, viz. (a) the ON-state loss (P_{on}); (b) the OFF-state loss (P_{off}) and (c) the switching loss ($P_{\text{switching}}$). The total power dissipation (P_{d}) is given as

$$P_{\text{d}} = P_{\text{on}} + P_{\text{off}} + P_{\text{switching}} \quad (1.5)$$

P_{on} is the average power loss when the diode is in the ON-state. One can use the piece-wise linear model [see Figure 1.4(d)] to estimate the power loss. The power loss is estimated as follows:

$$P_{\text{on}} = \frac{1}{T} \int_0^T v \times i \times dt \quad (1.6)$$

From the piece-wise linear model of the diode depicted in Figure 1.4(d), it can be observed that

$$v = V_{\text{d}} + (i \times r_{\text{d}}) \quad (1.7)$$

where V_{d} is the cut-in voltage or the knee voltage of the diode as shown in Figure 1.4(d). From Eqs. (1.6) and (1.7), the ON-state power loss in the diode is given as

$$P_{\text{on}} = (V_{\text{d}} \times I_{\text{Fav}}) + (I_{\text{Frms}}^2 \times r_{\text{d}}) \quad (1.8)$$

The knowledge of the average and rms values of the current in the circuit can be used to evaluate the ON-state conduction losses. OFF-state losses are due to the flow of the reverse saturation current

in the reverse-biased p – n junction. The reverse saturation current is negligibly small and therefore the i^2R loss due to it is generally not significant. The reverse recovery losses are negligible at low frequencies (e.g., at 50 Hz operation). However, at higher switching frequencies the reverse recovery losses are significant and affect the efficiency of the circuit considerably. In Figure 1.5(c), the shaded area represents the reverse recovery charge Q_{rr} . The loss during the reverse recovery is given as follows:

$$P_{\text{switching}} = E_{\text{switching}} \times f_s \quad (1.9)$$

where $E_{\text{switching}}$ is the energy spent during the reverse recovery process and f_s is the switching frequency.

$$P_{\text{switching}} = \left(\frac{1}{2} Q_{rr} V_R \right) \times f_s \quad (1.10)$$

Q_{rr} can be obtained from the datasheets and can be used to evaluate $P_{\text{switching}}$. If the shaded portion of Figure 1.5(c) is approximated as a triangle, then Q_{rr} can be estimated from the reverse current during turn-OFF I_{rr} as

$$Q_{rr} = \frac{1}{2} t_{rr} I_{rr} \quad (1.11)$$

Substituting Eq. (1.11) in Eq. (1.10), the switching loss can also be expressed as

$$P_{\text{switching}} = \frac{1}{4} I_{rr} V_R t_{rr} f_s \quad (1.12)$$

I_{rr} , V_R and f_s are circuit parameters which are known for the specific circuit. t_{rr} is obtained from the datasheet for the diode used. It is important to note that the switching losses are a function of the switching frequency. The junction temperature of the device should not exceed a certain limit (150°C as specified in the data-sheet). The heat generated in the junction should be removed in such a manner that at thermal equilibrium the junction temperature is well below the rated junction temperature. This is made possible by mounting the device on a heat sink. The thermal issues and the selection of heat sink for a specific calculated device power dissipation are discussed in Chapter 13.

Diode Model

A true model of a diode should take care of the steady-state and transient behaviors. One of the commonly used models for simulation is shown in Figure 1.6. Referring to Figure 1.6, i_d – V_d variables are characterized by an exponential relationship in accordance with Eq. (1.3). R_s is the Ohmic resistance of bulk material plus

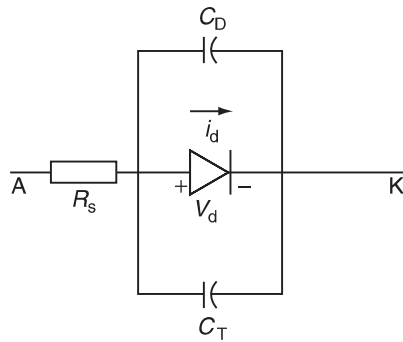


Figure 1.6 | Diode model.

the contact resistance. It is ideally zero; however, in practice it has a finite but small value. When a diode is abruptly switched from the forward to the reverse bias, the excess mobile charge has to be removed. This can be modeled as a diffusion capacitance C_D . For a transition from reverse to forward bias, the mobile charge must build up. This is characterized by a transition capacitance C_T . It should be noted that both C_D and C_T are capacitances whose values are not constant but dependent on the charge in it. Circuit simulator programs also use a similar model.

1.5 Bipolar Power Transistors

The bipolar transistor (bipolar junction transistor, BJT) is a three-terminal device. It has an emitter lead E, a collector lead C and a base lead B. The flow of current from the collector to the emitter is controlled by the current through the base. This base current is a fraction of the current through the collector. Therefore the BJT is called a current-controlled device. In the ON-state, the BJT can allow the flow of current in only one direction (from collector to emitter for NPN transistors; from emitter to collector for PNP transistors). Further, they can support only unidirectional voltages during the OFF-state.

BJT is a dual junction device with two possible configurations, namely, NPN or PNP. Figure 1.7(a) shows a schematic of the vertical cross-sectional structure of the NPN transistor. Figure 1.7(b) shows the circuit symbol of an NPN transistor. Referring to Figure 1.7(a), the transistor consists of a highly doped n -type emitter. The base region is a p -type doped semiconductor. The bottom layer is the n -type collector. The doping of the n -type collector is light towards the collector–base junction and higher towards the collector terminal. Generally, the collector is the largest region. Larger the collector region, greater is the amount of voltage that the transistor can withstand during the OFF-state. The arrow in the NPN transistor symbol points out of the emitter. In the case of the PNP transistor, the arrow in the emitter would point inwards as depicted in Figure 1.7(c). In general, the arrow would indicate the conventional direction of the current flow in the device. In the discussions to follow, NPN transistors only will be considered as the concepts of the NPN transistors can be extended to the PNP transistors without loss of generality.

Figure 1.8 depicts the basic operation of an NPN transistor. The base–emitter junction is forward-biased by connecting an external voltage V_E as shown. Electrons are injected from the emitter into the base. They

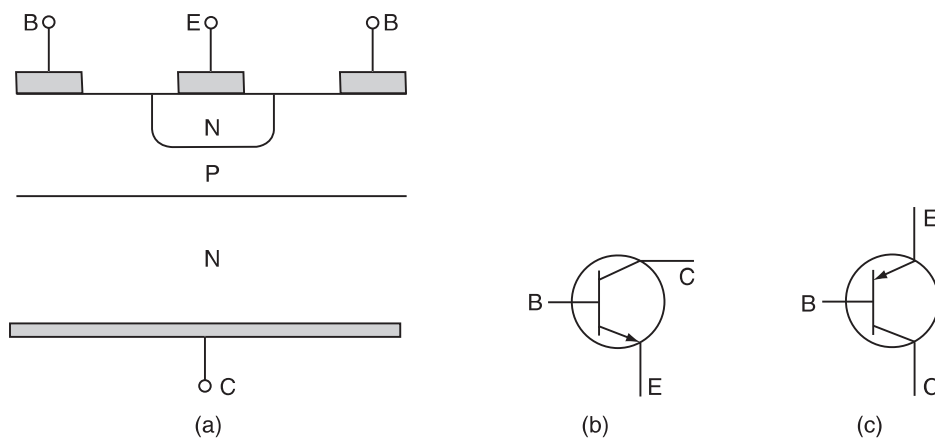


Figure 1.7 | NPN transistor: (a) Structure; (b) symbol (c) Symbol of PNP transistor.

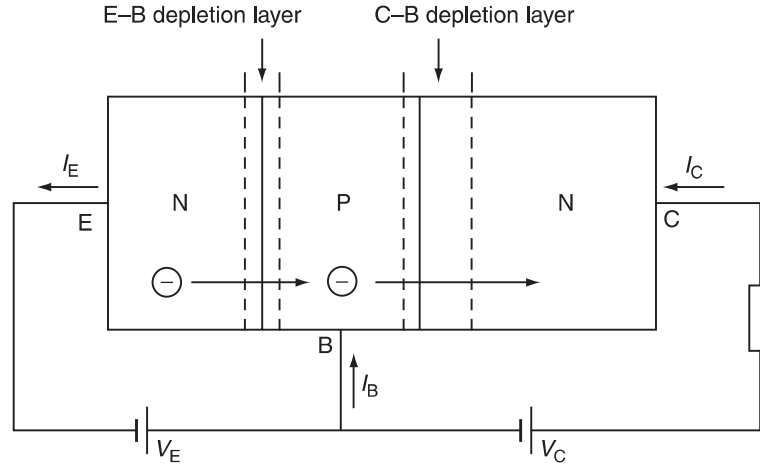


Figure 1.8 | Operation of NPN transistor.

appear at the emitter–base junction. Diffusion takes place in a manner similar to that of the p – n junction diode. However, the base is made intentionally small to prevent re-combination of electrons with holes in the p -region. This results in a very small I_B . As a result, the vast majority of electrons injected across the emitter–base junction move on to the base–collector junction and further into the n -material of the collector. Once the injected electrons appear in this region, the effect of V_C accelerates them towards collector, through the load resistor, through V_C and then back to V_E , thus completing the circuit.

The ratio of the collector current to the emitter current is called the “alpha parameter” of the transistor, that is, $I_C/I_E = \alpha$. This parameter varies between 0.9 and 0.98 depending on the base width. If the base were wide, then all electrons that have crossed the emitter–base junction will re-combine with holes in the base and there will be lesser or no electrons reaching the collector. In this case the emitter–base junction and collector–base junction will act as though they were diodes connected back-to-back. There is no transistor action taking place in such a case. From the discussion above, it should be noted that the transistor is not simply a back-to-back connection of two diodes.

The transistor connected as shown in Figure 1.8 is called the normal mode of operation and α is referred to as α_F or α_N . One can also have an inverted mode of connection wherein the collector acts as the emitter and vice versa. The α under this configuration is denoted as α_R or α_I . One can write the equivalent circuit model of the NPN transistor as shown in Figure 1.9. It is called the Ebers–Moll model. The equations for I_E and I_C for the normal and inverted mode of operations can be written as

$$I_C = \alpha_F I_F - I_{CO} (e^{V_C/V_T} - 1) \quad (1.15)$$

$$I_E = \alpha_R I_R - I_{EO} (e^{V_E/V_T} - 1) \quad (1.16)$$

Referring to the equivalent circuit model of the NPN transistor depicted in Figure 1.9,

$$I_F = I_{EO} (e^{V_E/V_T} - 1); \quad I_R = I_{CO} (e^{V_C/V_T} - 1); \quad I_C + I_E + I_B = 0$$

Here I_{EO} is the emitter–base junction saturation current when collector–base junction is zero-biased ($V_C = 0$); I_{CO} is the collector–base junction saturation current when emitter–base junction is zero-biased ($V_E = 0$).

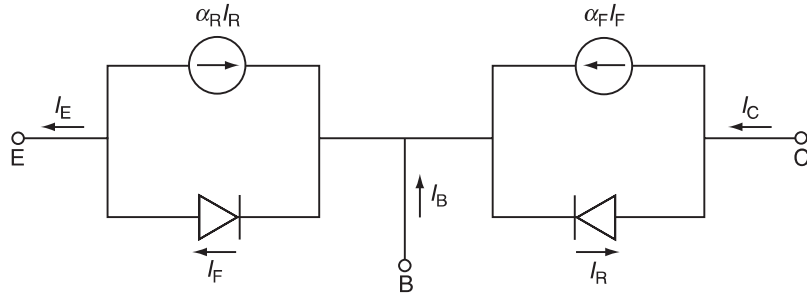


Figure 1.9 | Equivalent circuit of an NPN transistor.

Including the bulk resistance and capacitance (similar to the diode case) results in the model shown in Figure 1.10. C_{DE} and C_{TE} are the emitter–base junction diffusion and transition capacitances, respectively. Similarly, C_{DC} and C_{TC} are the collector–base junction diffusion and transition capacitances, respectively. Circuit simulation programs use a further refined version of the model called as the Gummel–Poon model.

Static Characteristics

BJT is a current-controlled device. Its output characteristics, that is, i_c versus V_{ce} is dependent on the base current i_b . Thus,

$$i_c = f(V_{ce}, i_b) \quad (1.17)$$

where V_{ce} is the collector-to-emitter voltage; i_b is the base current and i_c is the collector current.

Ideally $i_c = \alpha i_e = \beta i_b$ and should be independent of V_{ce} . However, due to “Early effect” (base width changes as V_{ce} changes) the parameter β changes with V_{ce} . Therefore, the i_c versus V_{ce} characteristics change with different values of the base current as depicted in Figure 1.11.

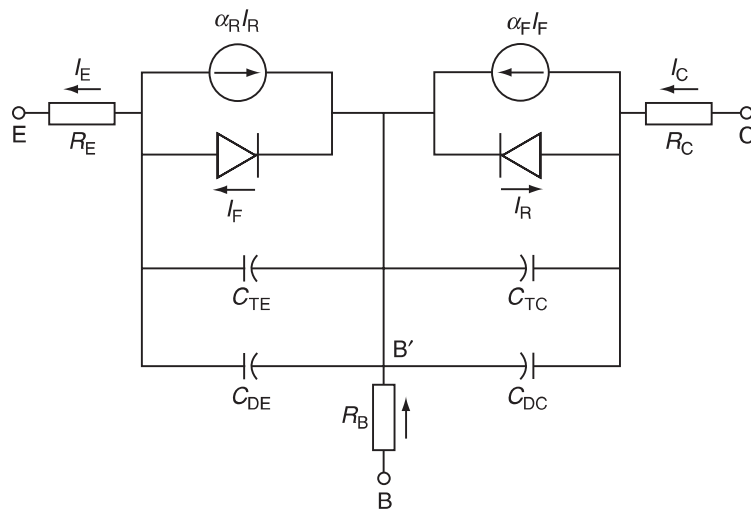


Figure 1.10 | Equivalent circuit of a transistor with bulk resistance and capacitance.

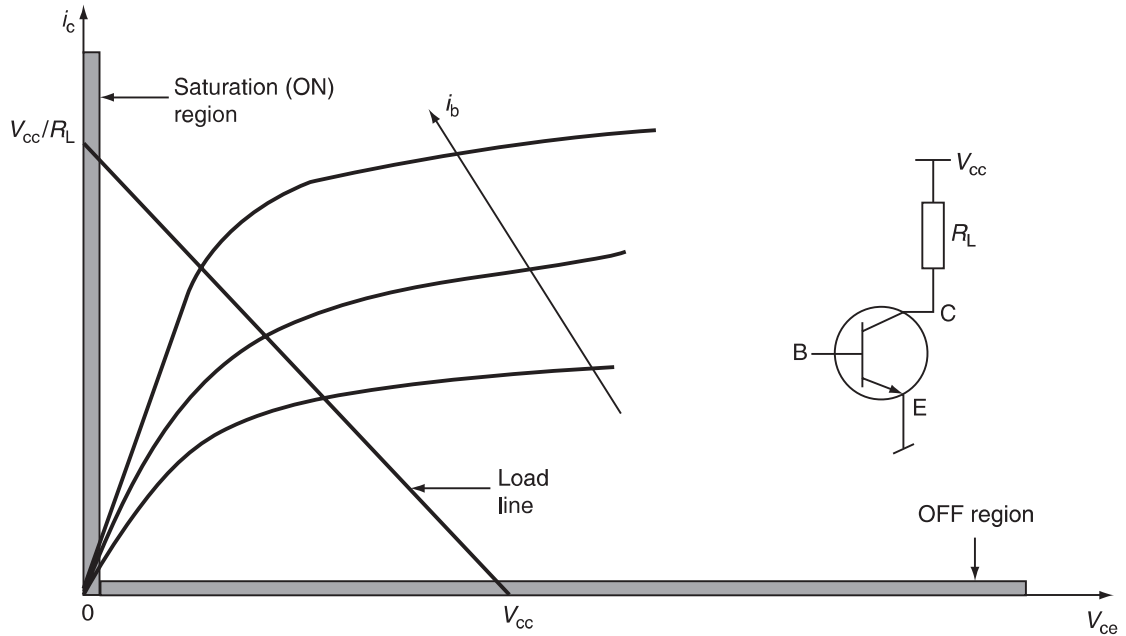


Figure 1.11 | Output characteristics of a transistor.

At very low V_{ce} values, the transistor is said to be in the ON-state. This region is called the saturation region. In this region, the collector current is independent of the base current and depends on the value of V_{ce} and load resistor R_L . A transistor parameter of interest is the ratio of the collector current to the base current, denoted by β or h_{FE} . The parameter h_{FE} is a useful number essential for design and is supplied by the manufacturers. Knowledge of i_C and h_{FE} gives the base current that will be needed to saturate the transistor or maintain the transistor in the ON-state.

At very low collector-current values, the transistor is said to be in the OFF-region as depicted in Figure 1.11. Here the transistor is capable of supporting the applied external voltage. The other region of the output characteristics that is not shaded is called the active region wherein the transistor behaves as a dynamic resistor. For a transistor to be operated as a switch, it is primarily switched between the saturation region or the ON-region and the OFF-region of the characteristics. However, during the transition from the saturation region to the OFF-region and back, the operating point may transit through the active region resulting in a loss called the switching loss.

Dynamic Characteristics

This sub-section describes the switching processes within a transistor under various base drive conditions. For a transistor that is in the ON-state, the following three charges may be said to exist.

1. A charge Q_b located in the base area.
2. A charge Q_{ce} located in the collector region underneath the emitter.
3. A charge Q_{cb} located in the collector region underneath the base contact.

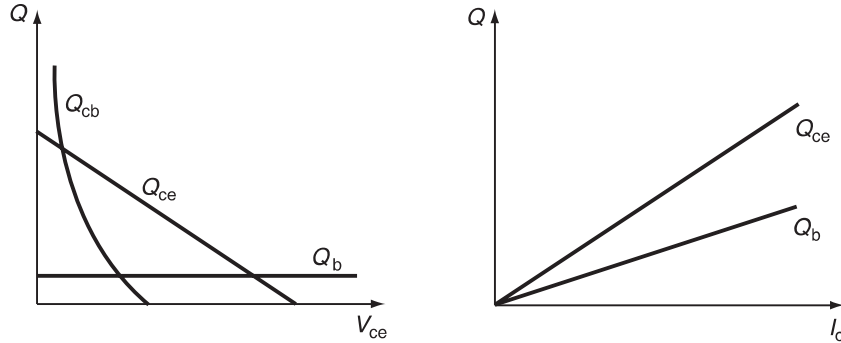


Figure 1.12 | Charge dependency on V_{ce} and I_c .

However, there are no charges present in a transistor that is in the OFF-state. Figure 1.12 illustrates the dependency of these charges with respect to the collector–emitter voltage and the collector current.

Referring to Figure 1.12, it may be observed that the charge Q_b is independent of the collector–emitter voltage drop. However, Q_b increases linearly as collector current increases. The charge Q_{ce} also increases linearly with the collector current but decreases linearly with increases in collector–emitter drop. To achieve lower collector–emitter drop, Q_{ce} must increase. The charge Q_{cb} that is located in the collector underneath the base contact has a significant effect on the collector–emitter voltage as depicted in Figure 1.12. The charge Q_{cb} increases rapidly as V_{ce} drop decreases and the transistor goes towards saturation.

Turn-OFF of Transistor

During turn-OFF, the charges in the transistor must be removed. For a transistor that is in the ON-state, the charges Q_b , Q_{ce} and Q_{cb} are as depicted in Figure 1.13(a). A collector current will flow only when there is a base charge Q_b . The charge Q_{ce} will be located in the collector region underneath the emitter. This results in a low Ohmic collector consequently making the collector–emitter voltage low. An increase in the charge, Q_{ce} results in a decrease in V_{ce} . The charge Q_{cb} will be located in the collector region underneath the base contact. When the base–collector region becomes forward-biased, Q_{ce} rapidly increases and V_{ce} decreases.

The process of turn-OFF of the transistor is illustrated in Figure 1.13. Referring to Figure 1.13(a), when the base voltage is made zero or negative, a negative base current flows as the collector potential is much greater than the base potential. This removes the charge Q_{cb} . Then the charge Q_c will be removed, starting from the area underneath the edges of the emitter as shown in Figure 1.13(b). Q_b also will now start to decrease.

As Q_b starts to decrease, the collector current will gradually be forced towards the center as shown in Figure 1.13(c). As long as Q_b is sufficiently high, the collector circuit will force the collector current to flow. However, V_{ce} will increase now as the current density becomes higher. With the emitter current reducing, the negative base-current flows through the base resistance underneath the emitter (R_b – base-spreading resistance). The charges Q_{ce} and Q_b now become increasingly located beneath the center of the emitter and thus have to be extracted through an increasing resistance. Consequently, the base–emitter terminal voltage becomes more negative. Subsequently, the emitter current is concentrated in the middle of the emitter. This marks the end of a period called the storage period wherein most of the stored charges in the base are removed. The storage period is denoted as storage time t_s and is in general given in the transistor datasheet.

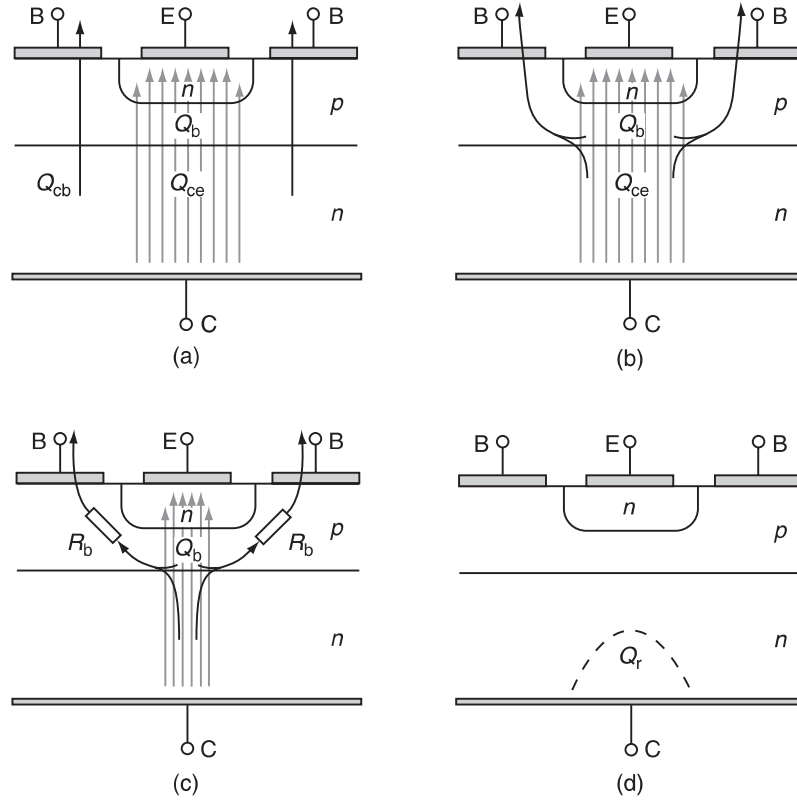


Figure 1.13 Turn-OFF of the transistor.

Fall time t_f begins as soon as Q_b is so low that the emitter injection of electrons starts to reduce. The emitter current decreases with a speed depending on the rate of decrease in Q_b . $I_e = 0$ when $Q_b = 0$. A trapped rest charge Q_r in the collector must still be removed by way of a collector–base current which appears as a tail current in the turn-OFF waveforms. This rest charge is indicated in Figure 1.13(d).

Turn-ON of Transistor

A transistor that is in the OFF-state has no charges within it. To obtain a low ON-state voltage, a base current is applied such that the collector charge Q_{ce} is built up, which will accordingly reduce the collector resistance and therefore the collector–emitter voltage (see Figure 1.12). To quickly build up the collector charge, the base-current waveform should have a peak at the beginning.

The dynamic characteristics of the transistor are shown in Figure 1.14. Referring to Figure 1.14(a), during the fall time t_f , the voltage across the collector–emitter [shown shaded in Figure 1.14(a)] is decided by the external circuit. If the load is resistive, then the voltage will rise linearly as the collector current falls linearly. If the load were an inductive one, then the voltage during the t_f period would be governed by $L di_L/dt$, where L is the inductance of the load and i_L is current through the load inductance.

Referring to Figure 1.14(b), during the rise time t_r , the current during this time [shown shaded in Figure 1.14(b)] is decided by the external circuit. If the load is resistive, then the current during this time will rise linearly. If the load across the collector–emitter of the transistor is capacitive, the current during the period t_r is governed by $C dV_{ce}/dt$, where C is the capacitance across the collector–emitter.

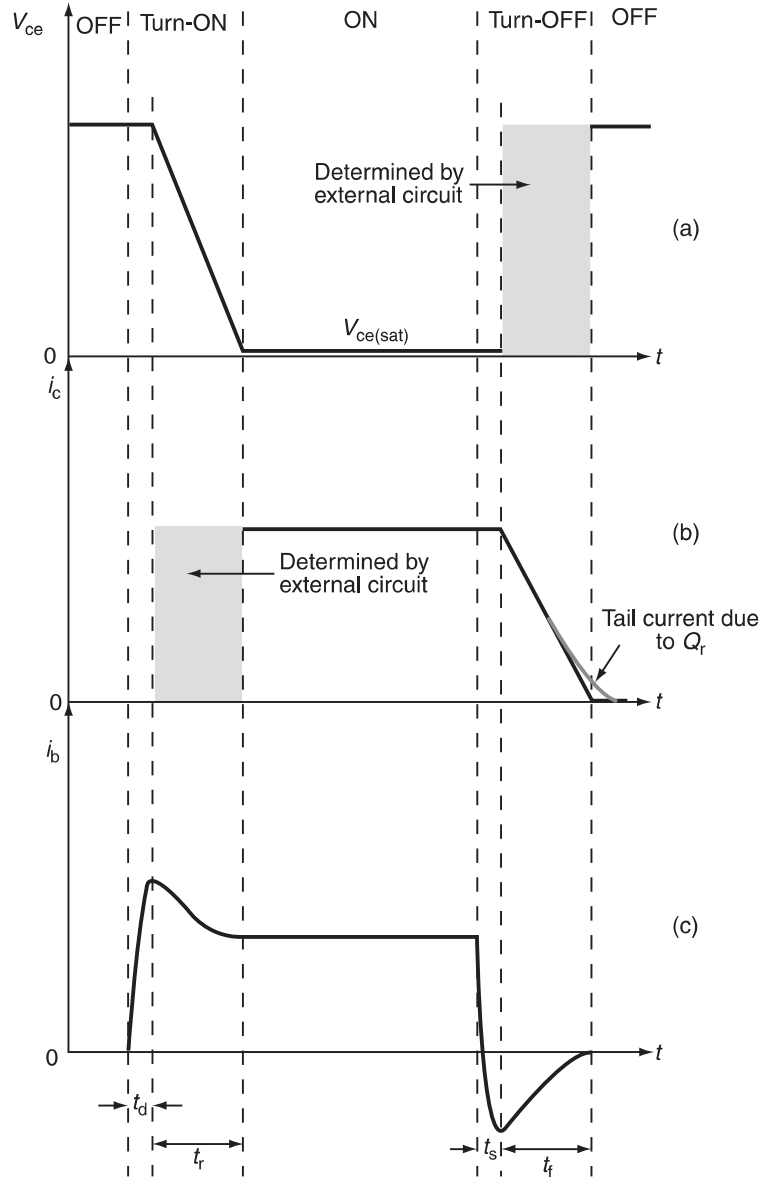


Figure 1.14 | Switching characteristics of a transistor: (a) Collector–emitter voltage waveform; (b) collector current; (c) base current.

Calculation of Power Dissipation in a Transistor

The different components of power loss in a transistor are: (a) the ON-state loss (P_{on}); (b) the OFF-state loss (P_{off}) and (c) the switching loss ($P_{switching}$). The total power dissipation (P_d) is given by

$$P_d = P_{on} + P_{off} + P_{switching} \quad (1.18)$$

P_{on} is the average power loss when the transistor is in the ON-state. The power loss is estimated as follows:

$$P_{\text{on}} = \frac{1}{T} \int_0^T v \times i \times dt \quad (1.19)$$

where v is the voltage across the collector–emitter of the transistor and i is the collector current through the transistor. During the ON-state, the voltage across the transistor, $v = V_{\text{ce(sat)}}$. Therefore Eq. (1.19) becomes

$$P_{\text{on}} = V_{\text{ce(sat)}} \times \frac{1}{T} \int_0^T i \times dt = V_{\text{ce(sat)}} I_{\text{cavg}} \quad (1.20)$$

For applications wherein the collector current is a pulsed current with a flat top I_c during the ON-state, the ON-state loss is given by

$$P_{\text{on}} = V_{\text{ce(sat)}} \times I_c \times D \quad (1.21)$$

where D is the duty cycle given by the ratio of the ON-time to the total switching period.

Like in the case of the diodes, the OFF-state losses are generally negligible. In the case of the switching losses, $P_{\text{switching}}$ depends on the nature of the load as is evident from the dynamic characteristics shown in Figure 1.14. A representative example case is discussed wherein the load is resistive. For resistive loads, assuming a linear rise and fall of voltages and currents, the switching losses can be calculated as discussed in the following sub-sections.

Switching Loss from OFF-State to ON-State

Referring to Figure 1.14, when the base drive signal is initiated, there is a time duration (t_d) called the delay time during which time the voltage and the current waveforms do not change much as the charges Q_b , Q_{ce} and Q_{cb} are being set up. This is followed by the time duration t_r . During the time t_r , the voltage across the device falls from V_{cc} to $V_{\text{ce(sat)}}$. Without much error, this fall may be assumed to be linear for a resistive load. The collector–emitter voltage during the rise time t_r is

$$v_{\text{ce}} = V_{\text{cc}} \left(1 - \frac{t}{t_r} \right) \quad (1.22)$$

The collector current for a resistive load will linearly rise from zero to I_c . This is given as

$$i_c = I_c \cdot \left(\frac{t}{t_r} \right) \quad (1.23)$$

The power dissipation during the OFF-state to ON-state transition is given by

$$P_{\text{OFF-ON}} = \int_0^{t_r} V_{\text{cc}} \left(1 - \frac{t}{t_r} \right) I_c \left(\frac{t}{t_r} \right) dt = \frac{V_{\text{cc}} I_c t_r f_s}{6} \quad (1.24)$$

where f_s is the switching frequency.

Switching Loss from ON-State to OFF-State

Referring to Figure 1.14, when the base drive signal is stopped, there is a time duration t_s called the storage time during which the voltage and the current waveforms do not change much as the charges Q_b , Q_{ce} and Q_{cb} are being removed. This is followed by the time duration t_f . During the time t_f , the current through the device falls from I_c to zero. Without much error, this fall may be assumed to be linear for a resistive load. The collector current during the fall time t_f is given as

$$i_c = I_c \left(1 - \frac{t}{t_f} \right) \quad (1.25)$$

The collector–emitter voltage for a resistive load will linearly rise from $V_{ce(sat)}$ to V_{cc} . This is given as

$$v_{ce} = V_{cc} \left(\frac{t}{t_f} \right) \quad (1.26)$$

The power dissipation during the ON-state to OFF-state transition is given by

$$P_{ON-OFF} = \int_0^{t_f} I_c \left(1 - \frac{t}{t_f} \right) V_{cc} \left(\frac{t}{t_f} \right) dt = \frac{V_{cc} I_c t_f f_s}{6} \quad (1.27)$$

where f_s is the switching frequency. The total switching loss is given as

$$P_{switching} = P_{OFF-ON} + P_{ON-OFF} \quad (1.28)$$

Substituting Eqs. (1.24) and (1.27) in Eq. (1.28), one obtains

$$P_{switching} = \frac{V_{cc} I_c f_s (t_r + t_f)}{6} \quad (1.29)$$

It should be observed that the switching losses are proportional to the frequency of switching. The load is generally never resistive. If the load is inductive, the voltage across the transistor during turn-OFF is determined by $L di_c/dt$, where L is the external load inductance. If the load is capacitive, the current through the transistor during turn-ON is determined by $C dV_{ce}/dt$, where C is the load capacitance as seen at the collector. The switching losses must be estimated for non-resistive loads also in a manner similar to that discussed for the resistive load. The total power dissipated is the sum of the ON-state loss and the switching loss. The power dissipated as above must be transported away from the junction such that the junction temperature remains at a safe value in equilibrium conditions. A proper heat sink has to be selected. The thermal calculations are discussed in Chapter 13.

1.6 Safe Operating Area (SOAR)

The maximum electric field across the collector–emitter of the transistor must remain below a critical value at all instants for proper functioning of the transistor. The electric field across the transistor is dependent on the collector-current density and the applied collector–emitter voltage. The electric field increases with increasing collector voltage. It also increases with increasing collector-current density. If the collector voltage is lowered then a higher collector-current density is permitted and vice versa. During switching transitions, there are some destructive combinations of the collector-current density and the collector voltage that are likely to occur. The SOARs give information about a given device on the current and voltage handling capabilities.

The collector-current density is dependent on the collector current and the amount of current crowding in the regions of the collector. The amount of current crowding is different for turn-ON (positive base voltage) and turn-OFF (negative base voltage) conditions. Therefore, the allowed combinations of the collector current and collector voltage will differ for turn-ON transition and turn-OFF transition. This information is available in the forward safe operating area (FSOAR) and the reverse safe operating area (RSOAR). A transistor in the ON-state or the OFF-state has the operating point along the Y -axis or the X -axis when either V_{ce} or i_c is zero. However, during switching transitions, both V_{ce} and i_c are non-zero and finite. This means that the operating point will be in the region of the I-quadrant of the $V_{ce}-i_c$ characteristic. In selecting a transistor, the one which operates within both the FSOAR and the RSOAR, for the specific circuit, should be chosen.

With positive voltage applied to the base, the shape of a typical SOAR characteristic is as shown in Figure 1.15(a). The solid line shows the SOAR for DC operation and the dashed lines shows the SOAR for pulsed operation. Operation outside the safe operating area is not allowed. For pulsed operation, the FSOAR increases

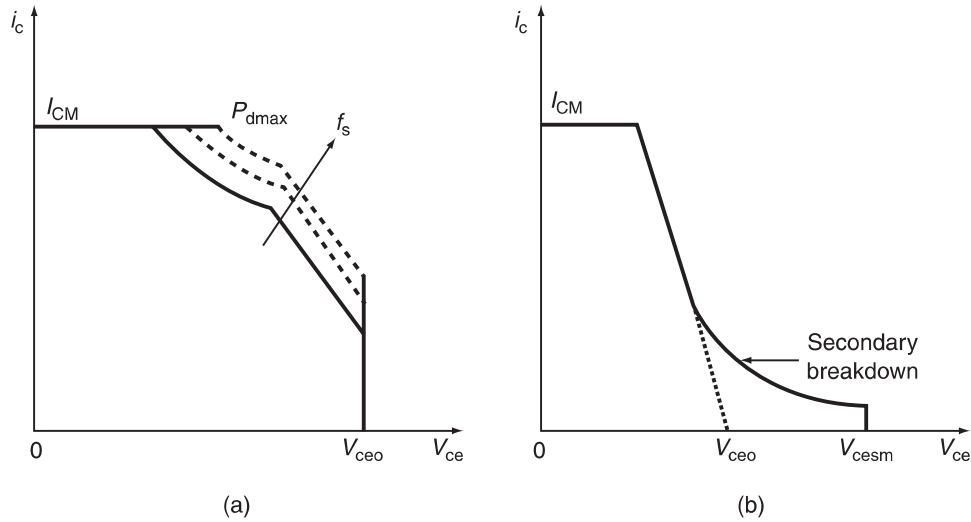


Figure 1.15 | Safe operating area (SOAR): (a) Forward SOAR (FSOAR); (b) reverse SOAR (RSOAR).

and for very small duty cycles, the FSOAR becomes square. The SOAR is designed to indicate the current, power dissipation, voltage and second breakdown limits of the transistor as depicted in Figure 1.15(a). The power dissipation limit is the hyperbolic portion of the limit curve that is given by the maximum possible dissipation allowed for the device. The second breakdown limit is the straight sloped line connecting the power dissipation limit and the voltage limit lines. The second breakdown is generally triggered by combinations of high collector-voltage and high collector-current density. With a positive voltage applied to the base, the region of highest current density is at the edge of the emitter which conducts a substantial proportion of the collector current. During sudden change of currents (especially during switching), a thermal gradient gets generated across the cross-section of the collector-emitter current flow. This thermal gradient will result in the non-uniform spreading of the current in the device. This uneven spreading produces localized hot spots and it can in turn reduce the local resistance, further increasing the non-uniformity. This will result in hot spots in the device that will finally destroy the device.

During turn-ON of the transistor, the high resistance of the collector region is reduced by the introduction of holes from the base and electrons from the emitter. This process is known as conductivity modulation. However, during turn-OFF of the transistor, these extra holes and electrons constitute a stored charge that must be removed from the collector before the voltage across the depletion region can develop. To turn OFF the transistor, a negative voltage is applied to the base and reverse base-current flows. During turn-OFF, it is essential that the device stays within its reverse safe operating area (RSOAR) that is shown in Figure 1.15(b).

1.7 Paralleling of Transistors

In high-power switching applications, transistors can be paralleled to share the load current. The transistor has a negative temperature co-efficient of resistance. As a result, one has to ensure that the current sharing between various parallel transistors is uniform, such that no transistor is over burdened and goes into a thermal runaway. It is very common to add an emitter resistance as shown in Figure 1.16 to equalize the currents. The emitter resistors give a negative feedback in the base-emitter circuit. If I_c increases then V_{be} will decrease, which would result in a decrease of I_b . This consequently will counteract the increase of I_c . The emitter resistor values are governed by the following equations:

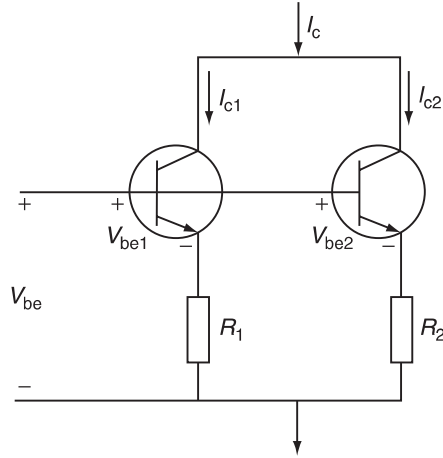


Figure 1.16 | Paralleling of transistors.

$$V_{be} = V_{be1} + RI_{c1} = V_{be2} + RI_{c2} \quad (1.30)$$

where $R_1 = R_2 = R$. From Eq. (1.30), one obtains

$$\Delta V_{be} = V_{be1} - V_{be2} = R\Delta I_c \quad (1.31)$$

Limit ΔV_{be} to around 0.2 V. Let ΔI_c (difference in collector currents between the two transistors) of about 0.5 A be tolerated. This means $R = 0.2/0.5 = 0.4 \, \Omega$. It should be noted that R dissipates power. Higher R means more dissipation and higher V_{be} (and V_{cc}) requirements. Paralleling transistors simplify heat-transfer problem by spreading the heat produced in a group of transistors over a large surface area heat sink.

1.8 Darlington Connection

High-power, high-voltage transistors generally have a low h_{FE} (β). The h_{FE} of saturated high-voltage transistors can be as low as 2. This means a very high base drive current is required. Therefore, the base drive circuit is no longer a simple low-power circuit. To circumvent this problem, Darlington connection as shown in Figure 1.17 can be used. In this case, the I_b requirement can be cut down to $I_c/(h_{FE} * h_{FE})$. It should

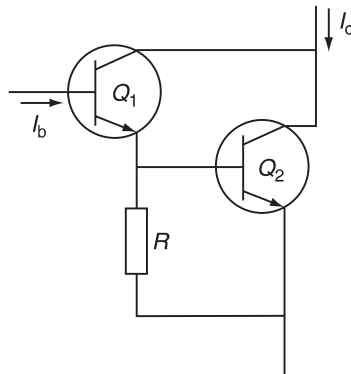


Figure 1.17 | Darlington connections of transistors.

be noted that $V_{ce(sat)}$ of the Darlington device is higher (~ 1 V to 1.2 V) as opposed to 0.4–0.6 V $V_{ce(sat)}$ of a single transistor. This implies that the power dissipation is higher in Q_2 for a given load.

1.9 Power MOSFETs

The metal oxide semiconductor field effect transistor (MOSFET) is a three-terminal device. It has a source lead S (analogous to the emitter of the transistor), a drain lead D (analogous to the collector of the transistor) and a gate lead G (analogous to the base of the transistor). The flow of current from the drain to the source is controlled by the voltage applied between the gate–source terminals. Thus the MOSFET is a voltage-controlled device. In the ON-state, the MOSFET can allow current in only one direction (from drain to source for n -channel MOSFETs and from source to drain for p -channel MOSFETs). In the OFF-state, they can support only unidirectional voltages in a manner similar to the BJT.

A structural schematic of an n -channel power MOSFET is shown in Figure 1.18(a). It consists of three layers, viz., (a) an n -type semiconductor that is connected to the drain D; (b) an n -type semiconductor that is connected to the source S and (c) in between the drain and the source is a p -type semiconductor that is on the substrate. Generally the substrate is connected to the source within the MOSFET. The gate is connected to a metallic conductor. This is insulated from the bulk of the MOSFET by an insulator that is generally an oxide of some metal such as silicon dioxide. The symbol for the n -channel MOSFET is shown in Figure 1.18(b) and the symbol for the p -channel MOSFET is shown in Figure 1.18(c). Note that for the n -channel MOSFET, the arrow mark shown in the symbol points inwards towards the gate and in the case of a p -channel MOSFET, the arrow mark points outwards, that is, away from the gate.

When a positive voltage is applied at the gate as shown in Figure 1.19, an electric field will be generated across the insulating oxide layer. This brings about a polarization of charges within the oxide layer. Consequently, these charges attract electrons from the p -material thereby creating an induced n -channel that

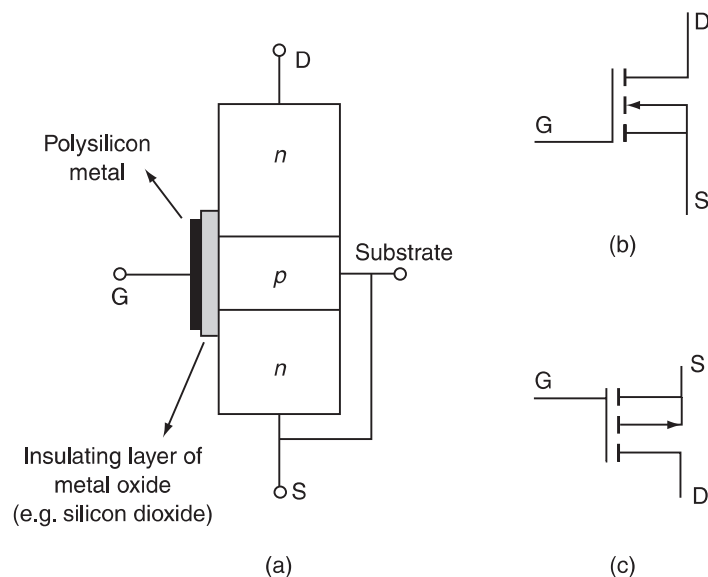


Figure 1.18 (a) Structural schematic of an n -channel MOSFET; (b) symbol of an n -channel MOSFET; (c) symbol of a p -channel MOSFET.

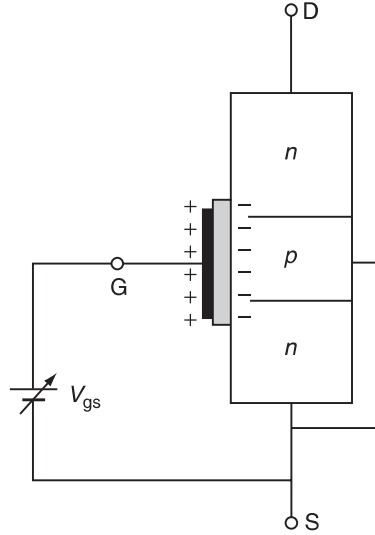


Figure 1.19 | Operation of the MOSFET.

bridges the n -type drain and source regions of the MOSFET. If a positive drain voltage is now applied between D and S terminals, the n -channel bridge provides a conducting path between the two n -regions. This conducting path that bridges the two n -type semiconductors in the MOSFET is called the **inversion layer**. As the voltage V_{gs} is increased, the width of the inversion layer increases. The MOSFET is said to have reached full enhancement when the inversion layer width is maximum.

It can be noted that there is a parasitic NPN BJT between the drain and source contacts with p substrate serving as the base of the parasitic BJT. To minimize the possibility that this transistor is ever turned ON, the p -type substrate is shorted to the source region as indicated in Figure 1.18. As a result of this short, a parasitic diode called the **body diode** exists between the drain and source of the MOSFET. This integral **body diode** can be used in many circuit configurations to advantage, thereby avoiding the use of an external diode that is required in such topologies.

Figure 1.18(a) gives the structural schematic of the MOSFET from a functional viewpoint. However, a typical cross-section of the practical MOSFET is as shown in Figure 1.20. The MOSFET has associated capacitances with respect to its terminals. This parasitic capacitance model of the MOSFET is shown in Figure 1.21. One should be aware that while manufacturing the MOSFET, a reverse diode from source to drain (i.e., anode at source and cathode at drain) gets inherently built by shorting the substrate and the source. This internal diode or the body diode is advantageous as it can be used for freewheeling purposes in inverters that are driving inductance loads.

Referring to Figure 1.20, the capacitance between the drain and the source C_{ds} varies in accordance with the width of the depletion layer that in turn depends on the voltage being supported by the device. The gate source capacitance consists of three components C_{gsn} , C_{gsp} and C_{gsm} (the metal to gate lead capacitance that is not indicated in Figure 1.20). Of these capacitances, C_{gsp} depends on the width of the inversion layer that in turn depends on the applied gate–source voltage. Of particular interest is the feedback capacitance C_{gd} . This capacitance plays a dominant role during switching. This is also the most voltage dependent. C_{gd} is essentially two capacitors in series such that

$$\frac{1}{C_{gd}} = \frac{1}{C_{gd\text{ox}}} + \frac{1}{C_{gd\text{bulk}}} \quad (1.32)$$

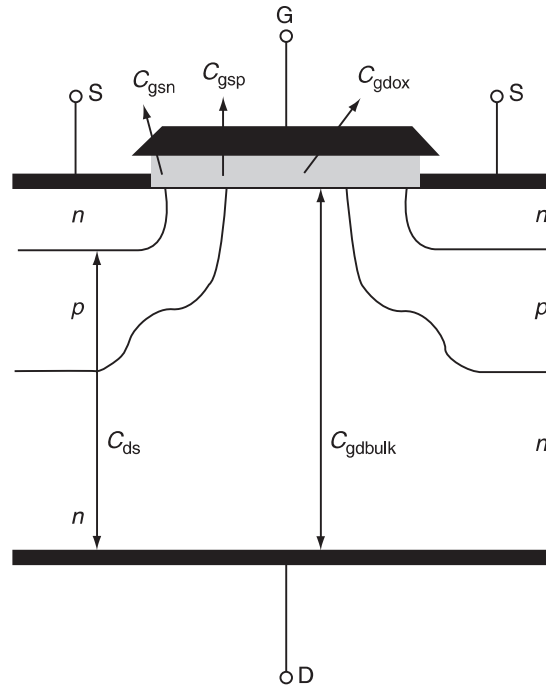


Figure 1.20 | Structure of the MOSFET.

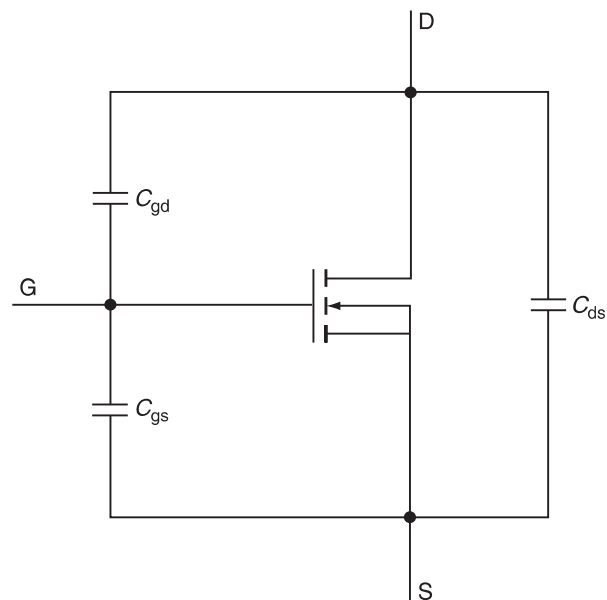


Figure 1.21 | Parasitic capacitance model of the MOSFET.

All the capacitances vary according to the thickness of the depletion region. Figure 1.21 depicts the capacitance model of the MOSFET. Most MOSFET datasheets do not refer to these capacitances. However, they provide information on the input capacitance C_{iss} , the output capacitance C_{oss} and the feedback capacitance C_{rss} . The datasheet capacitances relate to the parasitic capacitances shown in Figure 1.21 as follows:

C_{iss} : Parallel combination of C_{gs} and C_{gd}

C_{oss} : Parallel combination of C_{ds} and C_{gd}

C_{rss} : Equivalent to C_{gd}

Static Characteristics

The magnitude of the gate–source voltage V_{gs} essentially determines the drain current. Figure 1.22(a) shows the drain current i_d versus gate–source voltage V_{gs} characteristics. It can be observed from Figure 1.22(a) that there is a threshold voltage V_{gsT} below which the device is OFF. This V_{gsT} is of the order of 3–4 V in most power MOSFETs. Figure 1.22(b) shows the drain current i_d versus drain–source voltage V_{ds} . In the active region, the drain current is independent of the drain–source voltage and depends only on V_{gs} . This relation is approximately given by

$$i_d \propto (V_{gs} - V_{gsT})^2 \quad (1.33)$$

The MOSFET is said to be in the Ohmic region if V_{ds} is less than $(V_{gs} - V_{gsT})$. The boundary between the Ohmic and active region is depicted in Figure 1.22(b). The Ohmic region corresponds to the saturation region of BJT output characteristics. The MOSFET in the ON-state will operate in the Ohmic region. The drain to source resistance of the MOSFET in the Ohmic region is denoted by $R_{DS(ON)}$ and is an important selection parameter for the device.

In order to reduce the power dissipation in the ON-state, the device should have a low $R_{DS(ON)}$. A V_{gs} of ~10 V will take the MOSFET to the ON-state. However, in practical circuits, a V_{gs} of 15 V is applied in order to take the MOSFET to full enhancement such that the $R_{DS(ON)}$ is low. The maximum V_{gs} that can be applied is limited to +20 V, as only this much V_{gs} can be supported by the thin oxide layer.

Dynamic Characteristics

Power MOSFETs are intrinsically faster than BJTs as they have no excess minority carriers that must be moved into or out of the device during turn-ON and turn-OFF conditions. The only charges that must be

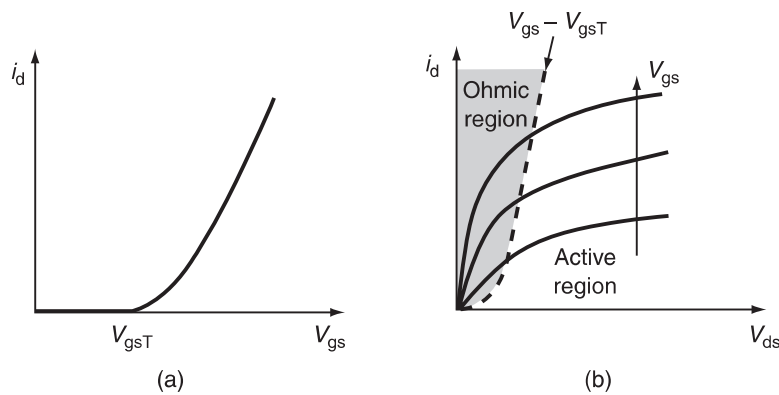


Figure 1.22 | Static characteristics of MOSFET.

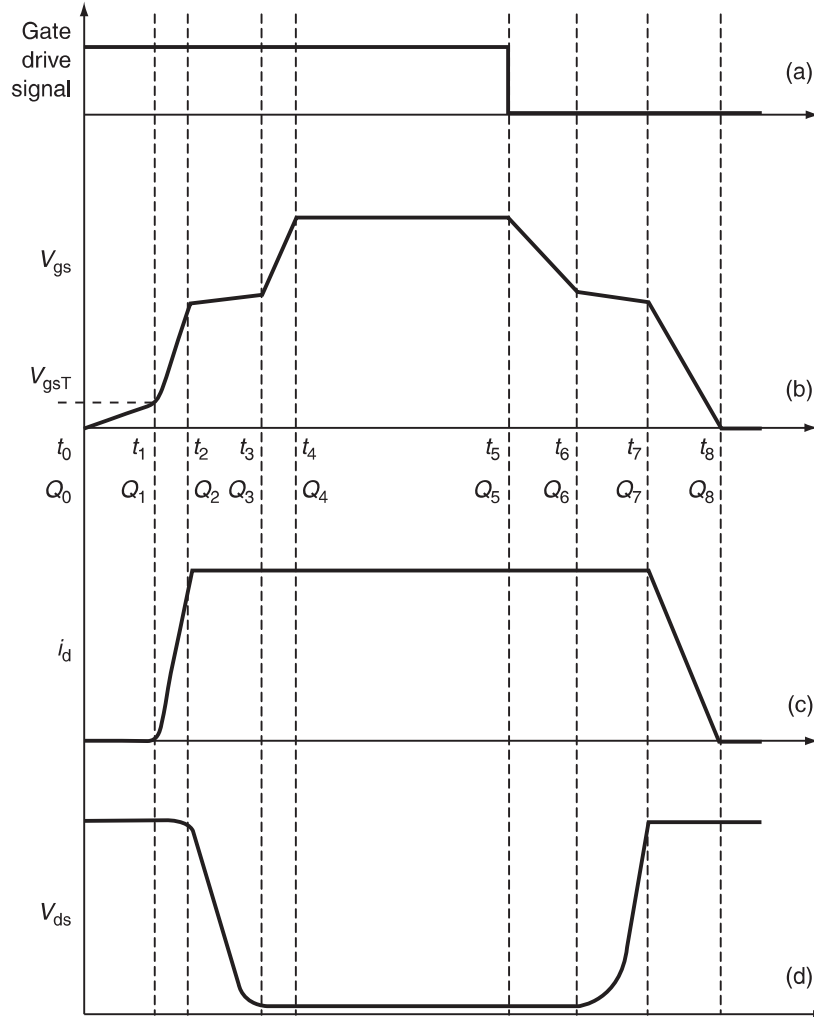


Figure 1.23 | Switching characteristic of MOSFET.

moved in or out are those on the stray parasitic capacitances due to oxide and depletion layers. The parasitic capacitances have been shown in Figure 1.21. The dynamic or the switching characteristics of the MOSFET are shown in Figure 1.23. The switching characteristic of the MOSFET is divided into various intervals. These intervals are explained based on the capacitance model of the MOSFET shown in Figure 1.21. For the purpose of the discussion regarding the switching behavior of the MOSFET, the current fed to the gate can be assumed to be a constant flowing into the gate terminal during the turn-ON period, that is, t_0 – t_4 . During the period t_4 – t_5 when the MOSFET is ON, the gate current is negligible. During the period t_5 – t_8 , when the MOSFET is turning OFF, the gate current is again assumed as constant but flowing out of the gate terminal. The equivalent circuit of the MOSFET during the turn-ON process at the various intervals is depicted in Figure 1.24. Referring to Figures 1.23 and 1.24, the switching action of the MOSFET is discussed referring to the various time intervals.

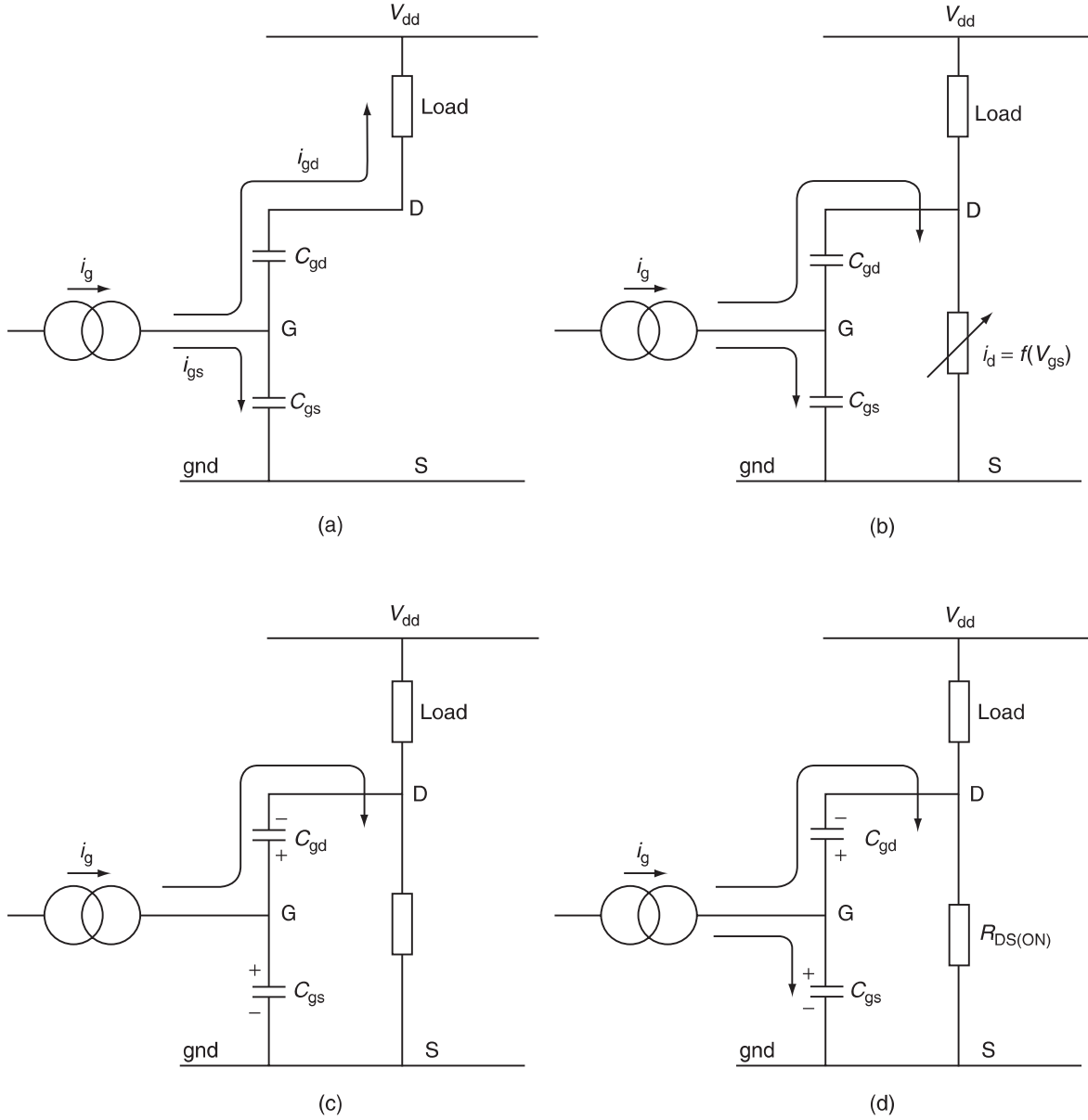


Figure 1.24 | Equivalent circuits during turning ON of a MOSFET: (a) During interval $t_0 - t_1$; (b) during interval $t_1 - t_2$; (c) during interval $t_2 - t_3$; (d) during interval $t_3 - t_4$.

During Interval $t_0 - t_1$

At t_0 , let there be an initial charge of Q_0 at the gate to source capacitance. An amount of charge ($Q_1 - Q_0$) is required to charge the gate to the threshold voltage V_{gsT} . The gate current flows through C_{gs} and C_{gd} as shown in Figure 1.24(a).

At turn-ON, C_{gs} is charged by a current

$$i_{gs} = C_{gs} \frac{dV_{gs}}{dt}$$

and C_{gd} is charged by a current

$$i_{gd} = C_{gd} \frac{dV_{gd}}{dt} \approx -C_{gd} \frac{dV_{ds}}{dt}.$$

During Interval t_1 – t_2

An amount of charge ($Q_2 - Q_1$) is required such that the full drain current i_d flows. The equivalent circuit is as shown in Figure 1.24(b).

During Interval t_2 – t_3

The equivalent circuit is as shown in Figure 1.24(c). During this interval, almost the entire gate current flows through the capacitance C_{gd} . As a consequence, the charge on C_{gs} is almost constant. This leads to an almost flat V_{gs} waveform during this interval as indicated in Figure 1.23(b). The drain-to-source voltage waveform starts falling as the MOSFET moves further into the Ohmic region. The rate at which V_{ds} falls as C_{gd} gets charged is

$$\frac{dV_{ds}}{dt} = -\frac{dV_{gd}}{dt} = \frac{i_g}{C_{gd}} \quad (1.34)$$

During Interval t_3 – t_4

Once the drain–source voltage has dropped to its ON-state value of $i_d R_{DS(ON)}$, C_{gs} gets charged such that the MOSFET is driven to full enhancement. An amount of charge ($Q_4 - Q_3$) is required for the MOSFET to be taken to full enhancement. The equivalent circuit is as shown in Figure 1.24(d).

During Interval t_4 – t_5

During this interval, the MOSFET is fully ON and no charge is required to maintain it in its ON-state. Thus the gate current is zero during the time when the MOSFET is fully ON.

During Interval t_5 – t_6

At instant t_5 , the gate signal to the MOSFET is made zero. Now the MOSFET begins the turn-OFF process. The charge ($Q_6 - Q_5$) is the excess gate charge from C_{gs} that is required to be removed to bring the MOSFET out of full enhancement.

During Interval t_6 – t_7

($Q_7 - Q_6$) amount of charge has to be discharged to bring the MOSFET out of the Ohmic region to the OFF-state. The drain–source voltage V_{ds} increases during this time as indicated in Figure 1.23. The feedback capacitance C_{gd} is discharged during this period. The rate at which the voltage across the feedback capacitance C_{gd} reduces will be same as the rate at which the drain–source voltage rises.

During Interval t_7 – t_8

($Q_8 - Q_7$) is the charge required to be discharged from C_{gs} to fully turn-OFF the MOSFET as indicated in Figure 1.23. One should note that the gate drive source should be able to sink during the discharge duration. The interval t_1 – t_2 where the drain current rises up to its full value corresponds to the rise time t_r and the interval t_7 – t_8 where the drain current falls from its full value to zero corresponds to the fall time t_f . [A typical MOSFET has the following specifications: for example, for BVZ 45 B, 500 V, 10 A MOSFET has $C_{gs} \sim 4000$ pF; $C_{gd} \sim 100$ pF; $C_{ds} \sim 300$ pF; $t_r \sim 100$ ns; $t_f \sim 120$ ns.]

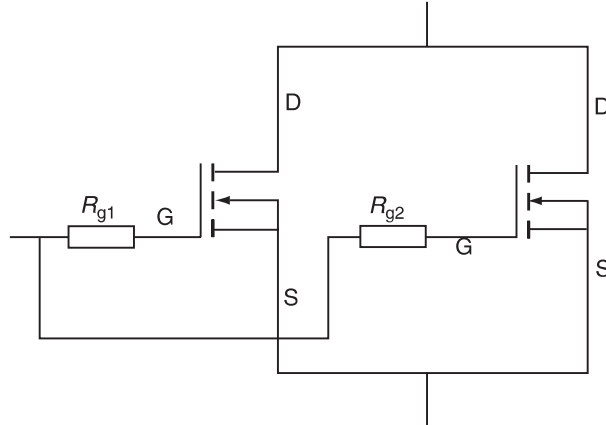


Figure 1.25 | Paralleling of MOSFETs.

Paralleling of MOSFETS

The Ohmic region resistance $R_{DS(ON)}$ is due to majority carriers. It has a positive temperature coefficient. This is advantageous when paralleling the devices. Current sharing resistors in series with the source are normally not required (as in the case of BJTs). Incidentally it can be noted that due to positive temperature coefficient of $R_{DS(ON)}$, the second breakdown will not take place in MOSFETs. In case there is a non-uniform distribution of the drain current, as the current increases in any section, the $R_{DS(ON)}$ will increase for that particular local section and as a consequence the current will reduce in that section and spread from the hot spot region to other areas.

If MOSFETs have to be connected in parallel, the schematic shown in Figure 1.25 is generally used. The drains and sources can be connected in parallel directly. While this can take care of the static condition, it is necessary to put small gate resistances of 10–100 Ω to damp high frequency oscillations that might result between gate leads and the parasitic device capacitances.

One should note that for a given rated current, the voltage drop will be much higher for a high-voltage MOSFET ($R_{DS(ON)} \sim 1$ to 2Ω) than for a low-voltage device ($R_{DS} \sim 0.1 \Omega$). This is due to the fact that the bulk n -region, that is, the drain region has to be larger in order to be capable of withstanding the electric field due to the higher applied voltage.

Handling Precautions for MOSFET

The gate–source oxide layer is very thin. Therefore, when touched with our body (that can have a minimum of 1–2 μC charge) the fields created can be extremely high so as to breakdown the oxide layer.

Remarks on BJT and MOSFET

1. BJTs have lower conduction losses in ON-state but higher switching losses whereas power MOSFETs have higher conduction losses and lower switching losses. Thus at higher frequencies, power MOSFETs are the obvious choice.
2. BJTs are current-controlled devices whereas MOSFETs are voltage-controlled devices.
3. It is easier to parallel power MOSFETs when compared to BJTs.
4. BJTs suffer from secondary breakdown, but the power MOSFETs do not suffer from secondary breakdown.

EXAMPLE 1.1

Consider a MOSFET with a $C_{gs} = 4000$ pf. From the basic charge–potential relationship,

$$Q = CV$$

The voltage across the gate–source capacitance that would get developed for a charge of $1\text{ }\mu\text{C}$ is

$$V_{gs} = \frac{Q}{C_{gs}} = \frac{1 \times 10^{-6}}{4 \times 10^{-9}} = 250\text{ V}$$

The maximum gate–source voltage that is allowable is only 20 V. Therefore, just touching a MOSFET may damage it permanently and hence the MOSFETs must be always carried in a conducting pad, so that the charge leaks if it tries to build up. While not in use, it is recommended to short the gate and source of the MOSFET with a conducting jumper or wire so that the gate–source voltage does not build due to spurious electrostatic fields. In order to limit the V_{gs} to 20 V, it is always a good practice to clamp the voltage across the gate–source terminals of a MOSFET by connecting a 15 V Zener diode close to gate and source leads.

Power Dissipated in a Power MOSFET

The ON-state loss for the MOSFET is given by

$$P_{ON} = I_{dms}^2 R_{DS(ON)} \quad (1.35)$$

$R_{DS(ON)}$ has a temperature coefficient of approximately 0.6% per °C and should be considered to calculate $R_{DS(ON)}$ (and P_{ON}) as the device temperature rises. In a manner similar to the discussion with regard to the BJT, the switching loss for resistive loads, assuming linear rise and decay of V_{ds} and i_d , is given as

$$P_{switching} = \frac{1}{6} V_{ds} I_d (t_r + t_f) f_s \quad (1.36)$$

Model for Power MOSFET

There are different levels of MOSFET models that have been used depending on the application. For integrated circuit fabrication using simulation, more complex models are necessary. Figure 1.26 shows a simple model that can be used for macro circuit simulation. In Figure 1.26, the capacitance subscript “gs” means gate to source, “gd” means gate to drain, “gb” means gate to bulk and “sb” means substrate to bulk.

1.10 Insulated Gate Bipolar Transistor (IGBT)

The IGBT is a combination of the advantageous features of MOSFET (like high drive input impedance, voltage control, fast switching) and BJTs (like low ON-state losses and high OFF-state voltage capability). The IGBT is also a three-terminal device. It has an emitter lead E (analogous to the emitter of

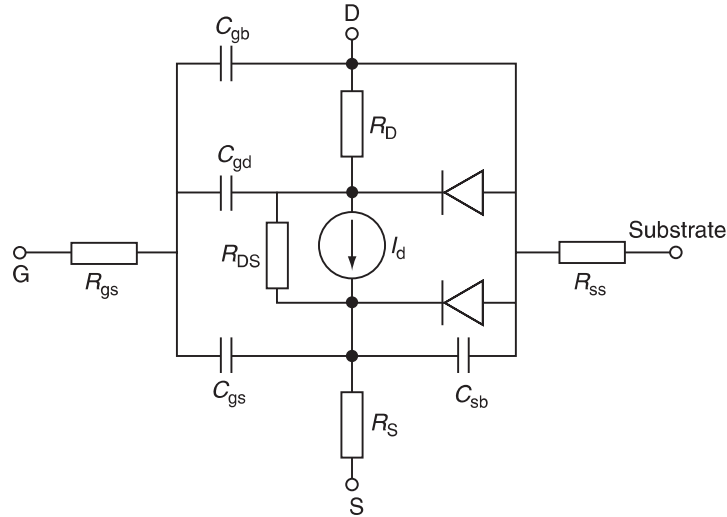


Figure 1.26 | Simulation model of a MOSFET.

the BJT), a collector lead C (analogous to the collector of the BJT) and a gate lead G (analogous to the gate of the MOSFET). The flow of current from the collector to the emitter is controlled by the voltage applied between the gate–emitter terminals. Therefore, like the MOSFET, the IGBT is also a voltage-controlled device. In the ON-state, the IGBT can allow current in only one direction (from collector to the emitter for *n*-channel IGBTs and from emitter to the collector for *p*-channel IGBTs). Further, they can support bipolar voltages during the OFF-state.

A structural schematic of an *n*-channel IGBT is depicted in Figure 1.27(a). It consists of four layers: a *p*-type semiconductor that is connected to the collector C, an *n*-type semiconductor that is connected to the emitter E and in between is a *p*–*n* semiconductor junction. The gate is connected to a metallic conductor. This is insulated from the bulk of the IGBT by an insulator that is generally an oxide of metal, like, silicon dioxide. The symbol for the *n*-channel IGBT is shown in Figure 1.27(b) and that for a *p*-channel IGBT is shown in Figure 1.27(c).

When a positive voltage is applied at the gate as indicated in Figure 1.28, an electric field will be directed through the insulating oxide layer. This brings about polarization of charges within the oxide layer. As a consequence, electrons from the *p*-material get attracted towards the insulating oxide layer thereby creating an induced *n*-channel that bridges the two *n*-type sections of the IGBT. The induced *n*-channel along with the two *n*-sections forms an equivalent *n*-type semiconductor. The *p*-type semiconductor connected to the collector C along with the equivalent *n*-type semiconductor forms an equivalent *p*–*n* junction and behaves like a diode. Because of the formation of the equivalent *p*–*n* junction, the IGBT is capable of withstanding **reverse** voltages also during the OFF-state.

From the structural schematic of the IGBT, it is evident that an IGBT can be viewed equivalently as a MOSFET with a diode connected to its drain as indicated in Figure 1.29. Alternatively, it can also be viewed as a MOSFET and a PNP transistor as shown in Figure 1.30.

The detailed structure of the IGBT is shown in Figure 1.31. One should note that the IGBT has a parasitic thyristor between its collector and emitter as indicated in Figures 1.31 and 1.32. This parasitic thyristor should not latch. If it latches then the IGBT will lose the gate control. Therefore, when the IGBT is

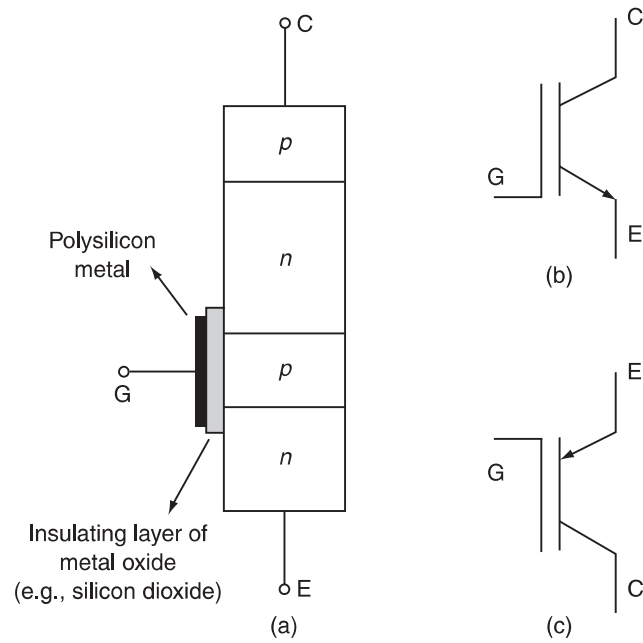


Figure 1.27 (a) Structural schematic of an n -channel IGBT; (b) symbol of an n -channel IGBT; (c) symbol of a p -channel IGBT.

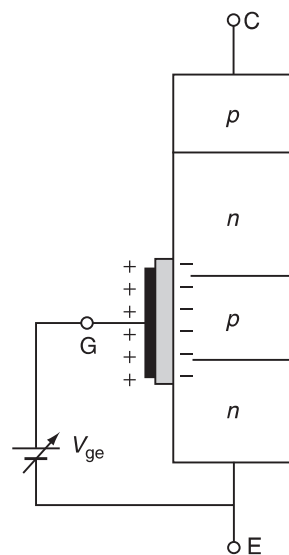


Figure 1.28 Operation of the IGBT.

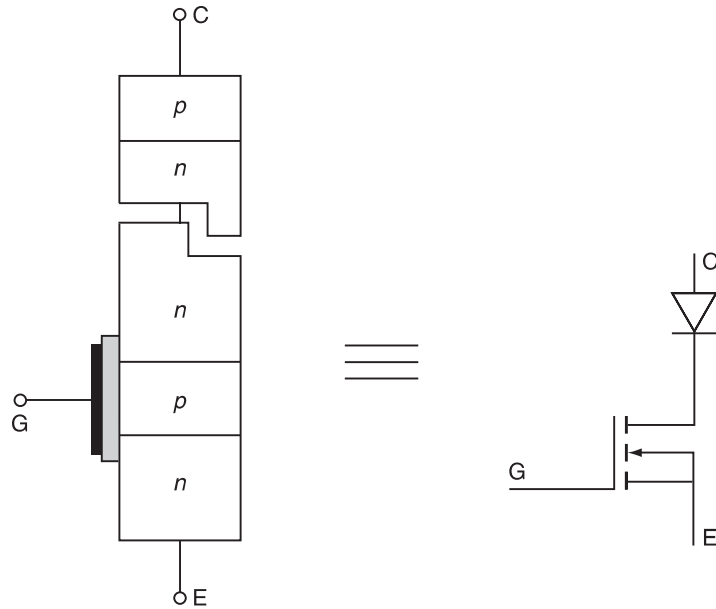


Figure 1.29 | Equivalent circuit of IGBT: Diode and MOSFET.

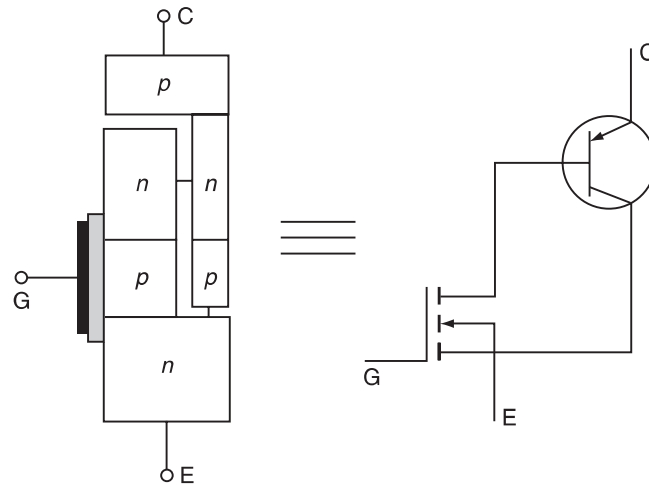


Figure 1.30 | Equivalent circuit of IGBT: Transistor and MOSFET.

manufactured, R_s as indicated in Figure 1.32 is made very low so that the drop across it is less than cut-in voltage of the base–emitter of the NPN transistor in the parasitic thyristor. This way the NPN transistor is always off, thereby avoiding the latching problem. One should be aware that while manufacturing the IGBT, a reverse diode from emitter to collector (i.e., anode at emitter and cathode at the collector) is built separately as it is not inherent like in the case of the MOSFETs. This internal diode or the body diode is required for freewheeling purposes in topologies that are driving inductance loads.

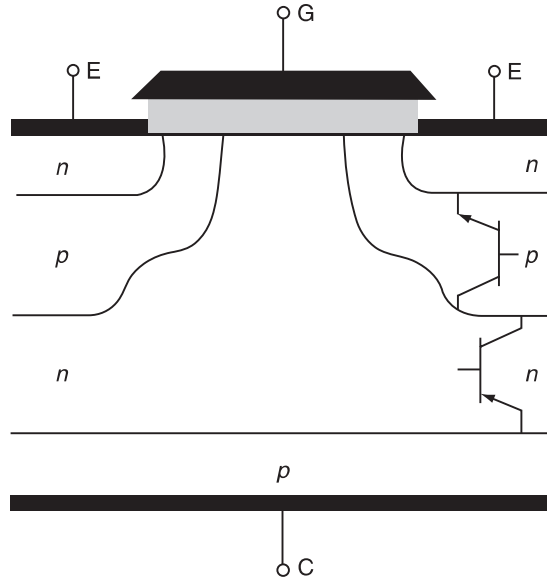


Figure 1.31 | Structure of an IGBT.

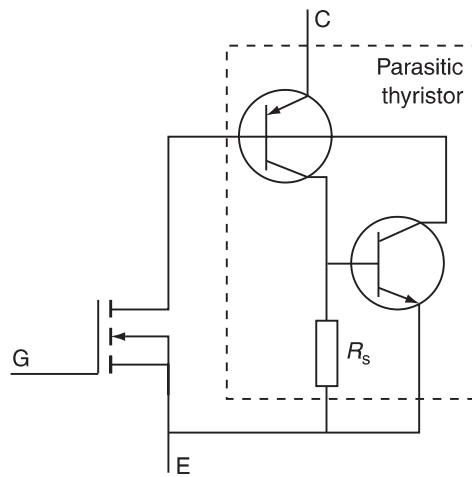


Figure 1.32 | Equivalent circuit of the IGBT showing the parasitic thyristor.

Static Characteristics

The $i_c - V_{ce}$ characteristics of an n -channel IGBT are shown in Figure 1.33 and the $i_c - V_{ge}$ characteristics are shown in Figure 1.34. From Figure 1.33 it is observed that the IGBT can support both forward and reverse voltages during the OFF-state. From Figure 1.34, it is seen that if V_{ge} is less than the threshold voltage, the IGBT is in the OFF-state.

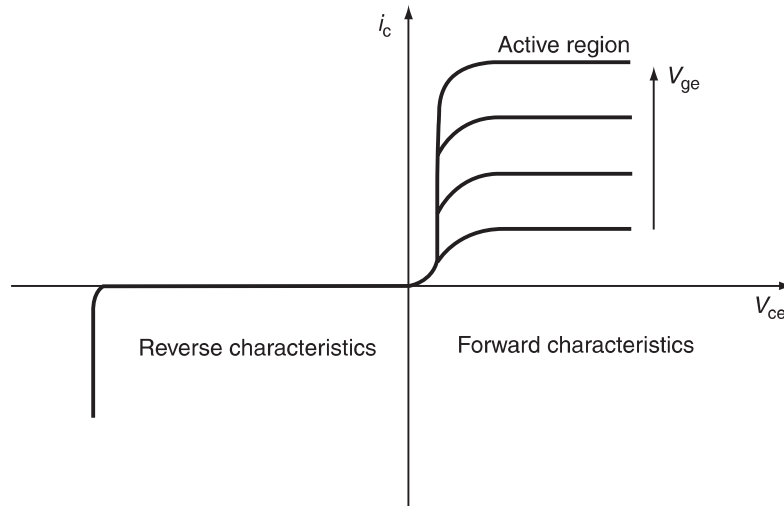


Figure 1.33 | Static i_c – V_{ce} characteristic of an IGBT.

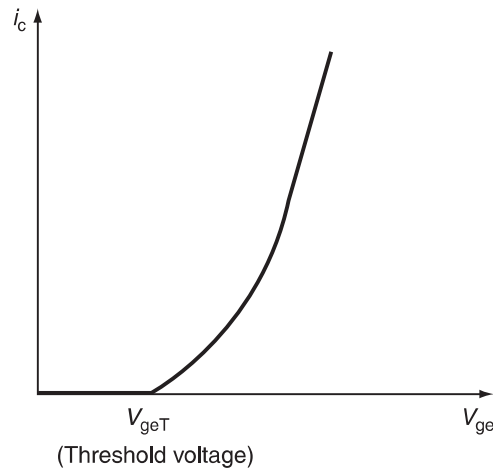


Figure 1.34 | Static i_c – V_{ge} characteristic of an IGBT.

In comparison, for a given i_c , the $V_{ce(sat)}$ of the IGBT is less than the V_{ds} of a comparable MOSFET when ON but greater than $V_{ce(sat)}$ of a comparable BJT.

Dynamic Characteristics

The dynamic behavior of an IGBT is similar to that of a MOSFET when viewed from the gate side and is similar to that of a BJT when viewed from the collector side. However, the maximum switching frequency is limited as compared to that of a MOSFET.

The switching times of the IGBT are related to the gate–emitter voltage and collector–current waveforms. Figure 1.35 shows the typical switching waveforms. The switching times of an IGBT are mainly determined by its internal capacitances and parasitic inductances together with the internal resistance

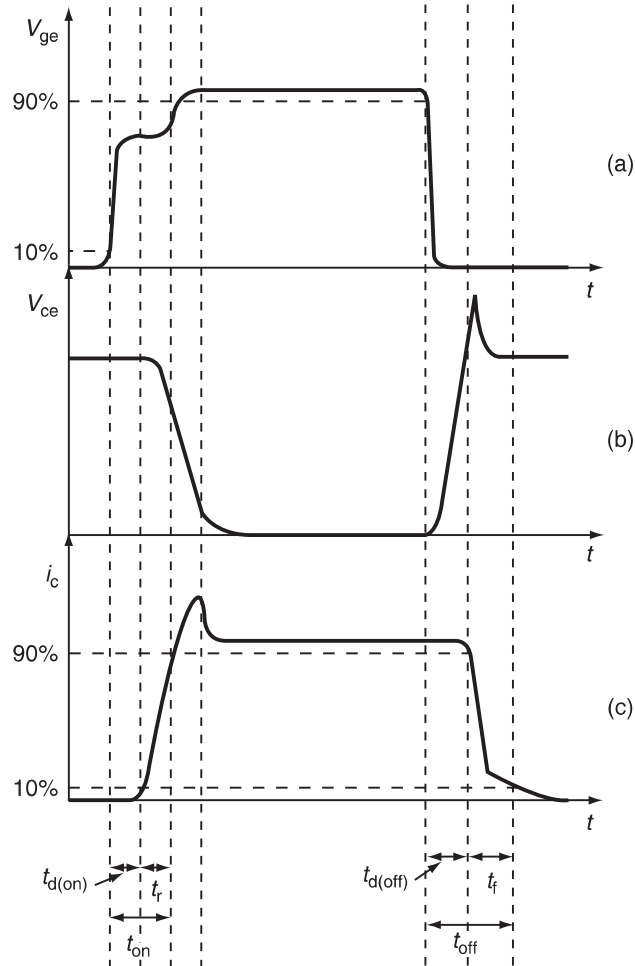


Figure 1.35 | Dynamic switching characteristics of an IGBT.

of the gate-side voltage source. In order to charge and discharge the capacitances rapidly and to reduce the transients caused by the gate circuit inductance, a low internal impedance of the gate-side voltage source is desirable. It shortens the switching times and reduces the switching losses. On the other hand, a very fast turn-ON causes a high peak reverse recovery current through the body diode that appears as an additional peak collector current. During very fast turn-OFF, a high transient voltage is caused by the parasitic collector–emitter inductance as shown in Figure 1.35(b).

It is very important to keep the parasitic inductance in the gate circuit at a minimum by using very short leads. This inductance might otherwise generate parasitic oscillations in conjunction with the IGBT capacitances. The maximum rated gate–emitter voltage as specified in the datasheets in most cases is ± 20 V. It is recommended to connect a 15 V Zener diode between the gate and the emitter as a protective clamper.

Turn-ON

During turn-ON, the IGBT behavior is more like that of a MOSFET. The collector–emitter voltage, the collector-current and the gate–emitter voltage waveforms are similar to the turn-ON waveforms of the

MOSFET. When the gate–emitter voltage reaches the threshold value of V_{geT} , the collector-current starts to rise. The time interval between the instant when V_{ge} reaches 10% of its final value and the instant when i_c reaches 10% of its final value is called the turn-ON delay time $t_{d(on)}$. The subsequent time interval up to the instant when the collector current reaches 90% of its final value is called the rise time t_r . During this period of time, most of the turn-ON power dissipation takes place. The sum of the turn-ON delay time $t_{d(on)}$ and the rise time t_r is called the turn-ON time t_{on} . At the end of t_{on} , the collector–emitter voltage V_{ce} often has not yet fallen to its final value of $V_{ce(sat)}$. This has to be considered when calculating the turn-ON dissipation. The collector-current peak shown in Figure 1.35 indicates the peak reverse recovery current of the body diode or the freewheeling diode. This peak current has to be taken into account in any turn-ON power dissipation calculations as well.

Turn-OFF

During turn-OFF, the IGBT behaves more like a BJT. To turn the IGBT OFF, the voltage in the gate control circuit is switched to zero. Referring to Figure 1.35, there is first a turn-OFF delay time $t_{d(off)}$ that is the interval between the instant when the gate–emitter voltage has fallen to 90% of its initial value and the instant when the collector current has fallen to 90% of its initial value. The subsequent period of time up to the instant when the collector current has fallen to 10% of its initial value is called the fall time t_f . The sum of $t_{d(off)}$ and t_f is called the turn-OFF time t_{off} .

Tail Current

The tail current and tail time are properties that are specific to the IGBT. However, they also depend on the operating conditions. The tail current is higher for lower saturation voltage $V_{ce(sat)}$. The trade-off for the favorable reduction of the conducting state power loss is an increase in the tail current. Since at pulse frequencies of 10–20 kHz the switching losses are more prevalent, most IGBTs are designed for a low tail current at the expense of the saturation voltage which would be a little higher than the minimum possible value.

Remarks on IGBT

1. The power loss calculations for the IGBT are similar to the loss calculations indicated for BJTs and MOSFETs.
2. The gate drive power is low such as in the case of MOSFETs.
3. IGBT has considerably greater overload capability as compared to a MOSFET. For a MOSFET, the peak overload is about 5–6 times the continuous drain current whereas for the IGBT, the peak overload is about 20 times the continuous collector current. An IGBT is capable of taking a short-circuit current (across a 600 V bus) for a period of 10 μ s.
4. Over voltage robustness is less than that of MOSFETs.
5. IGBTs are more like the BJTs during turn-OFF and like the MOSFETs during turn-ON.
6. Unlike the BJT, where the collector current is limited by the current gain, that is $(h_{FE(sat)}i_b)$, the IGBTs do not have such limitations. The collector current is limited only by the external source.

1.11 Thyristor Family

The term thyristor is a generic name for a semiconductor switch having four or more layers and is in essence a p – n – p – n structure. Thyristors form a large family of semiconductor switches. If an Ohmic connection is made to the first p -region and the last n -region and no other connection is made to any other intermediate region, then the device is a diode thyristor. If an additional Ohmic connection is made to the

intermediate n -region or the intermediate p -region, the device is called a triode thyristor. If an Ohmic connection is made to both intermediate regions, then the device is a tetrode thyristor. All such devices have a forward characteristic of the form shown in Figure 1.36.

There are three categories of thyristor reverse characteristic: blocking (as in normal diodes like in symmetric SCRs), conducting (large reverse current at low reverse voltages like in asymmetric SCRs and GTOs) and approximate mirror image of the forward characteristic (bi-directional thyristors like in DIACs and TRIACs). The simplest thyristor structure and the most common is the reverse blocking triode thyristor usually referred to as the silicon-controlled rectifier (SCR). The more complex thyristor structure is the bi-directional triode thyristor or TRIAC. In this section, the various devices in the thyristor family will be briefly reviewed.

Silicon-Controlled Rectifier (SCR)

SCR is a three-terminal device with an anode terminal A, a cathode terminal K and a control or gate terminal G. Structurally it is similar to a BJT with an extra p -layer at the anode. The structure of the SCR is depicted in Figure 1.37(a) and the circuit symbol is shown in Figure 1.37(b).

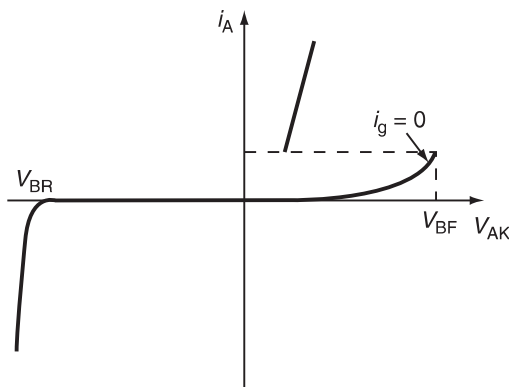


Figure 1.36 | Forward characteristics of the thyristor family.

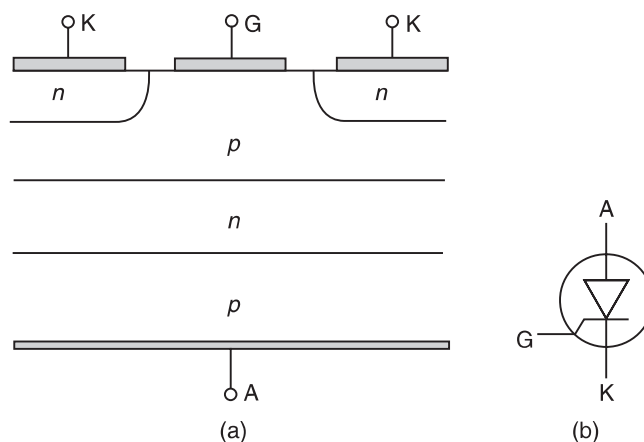


Figure 1.37 | (a) Structure of an SCR; (b) symbol of an SCR.

Static Characteristics

Application of a negative voltage to the anode–cathode of an SCR will reverse bias the SCR. In this condition, the junctions J_1 and J_3 are reverse-biased and J_2 is forward-biased. The device is said to be in the OFF-state and blocks the reverse voltage. On the other hand, applying a positive voltage to the anode–cathode, as shown in Figure 1.38, with the gate open, will forward bias the SCR. With the anode–cathode voltage positive, it can be observed that junctions J_1 and J_3 are forward-biased and J_2 is reverse-biased. The SCR is still in the OFF-state and blocks the applied forward voltage. This is a property unique to the thyristor family. If forward-bias voltage is further increased, the junction J_2 will breakdown and the device turns ON behaving in a manner similar to a diode. The static characteristic is shown in Figure 1.39. Figure 1.39(a) shows the static characteristics when the gate current is zero. The voltage at which the junction J_2 breaks down and the SCR conducts is called the forward breakover voltage V_{BF} . If the gate of the SCR is connected to a voltage source and a gate current is allowed to flow, then the breakover voltage reduces as indicated in Figure 1.39(b).

The SCR action is best understood by modeling it as two BJTs connected as shown in Figure 1.40. With the gate open or shorted to the cathode, the device is OFF and no current flows from anode to cathode, except for a very low leakage current. If an external positive current pulse is applied to the gate, it becomes the base current of the NPN transistor Q_2 . Consequently, the collector current of Q_2 supplies the base current to the PNP transistor Q_1 . The collector current of Q_1 then further increases the base current of Q_2 and so on. This iterative action maintains the device in the conducting state even if the gate signal is now removed. The device continues to conduct till the anode voltage is less positive than the cathode voltage. Figure 1.39(b) shows the static characteristics with gate current as a parameter. It should be noted that once the device is ON, the gate loses complete control of the device and the device cannot be turned OFF through the gate. The SCR can be switched OFF only if the anode current is brought below a threshold called the **holding current** value for that SCR.

Dynamic Characteristics

di/dt Effect On supplying the gate trigger to a forward-biased SCR, the SCR will turn ON. The anode current starts to rise after a small delay. The rate of change of the anode current will depend on the nature of

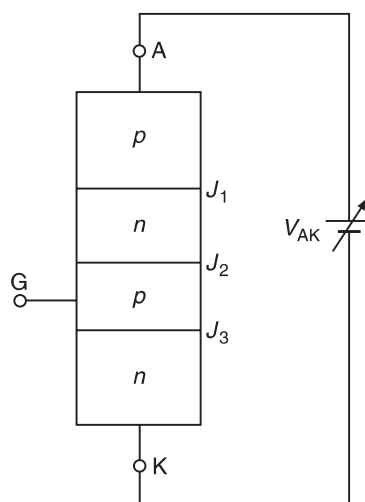


Figure 1.38 | Biasing the SCR.

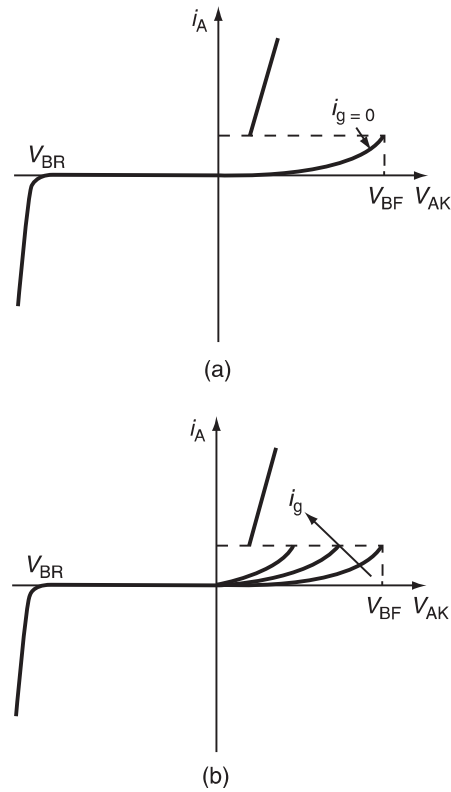


Figure 1.39 | V - I characteristic of an SCR (a) with gate open; (b) with gate current applied.

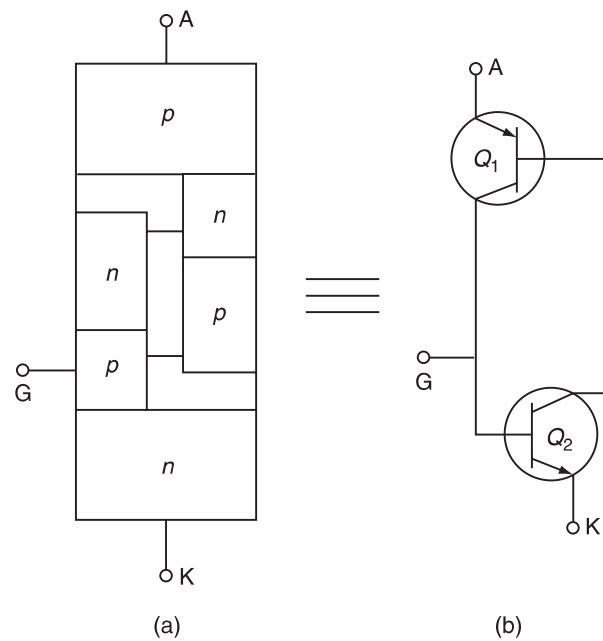


Figure 1.40 | Two transistor equivalent circuit of an SCR.

the load. Initially the ON-state current is concentrated in a small area around the gate region. If the rate of rise of this ON-state current is high, then the area around the gate region will become overheated that may permanently damage the device. Therefore, it should be ensured that the external circuit di/dt encountered by the SCR should be less than the rated di/dt of the device.

Turn-OFF Characteristics The turn-OFF characteristic waveforms of the anode current and the anode–cathode voltage are similar to that of the diode with the added restriction that all junctions including the gate junction should fully recover before the SCR gets into a forward-blocking state. The turn-OFF characteristics are depicted in Figure 1.41.

A high rate of rise of the applied forward voltage across the anode and cathode (dv/dt) can trigger the SCR. To keep the SCR turned OFF, the re-application of the OFF-state forward voltage must be delayed to avoid re-triggering due to high dv/dt . The circuit commutated recovery time t_q is measured from the zero crossover of the current as shown in Figure 1.41. It should be noted that t_q is an important parameter. SCRs are often classified as line-commutated or converter grade if $t_q > 50 \mu s$ and as inverter grade if $t_q < 20 \mu s$. For the line commutated converter, t_q is less important since the half-period is 10 ms. But for inverter circuits switching at higher frequencies, inverter grade SCRs with appropriately lower t_q will have to be used.

dv/dt Effect Figure 1.41 shows a possible problem with large rate of forward voltage, dv/dt . If dv/dt is large, the SCR may self-trigger even in the absence of gate current. This is due to the fact that there exists a leakage current through the reverse-biased junction. This leakage current flows through the capacitance of junction J_2 and is given by $i = c dv/dt$. This leakage current is dependent on dv/dt . If this leakage current

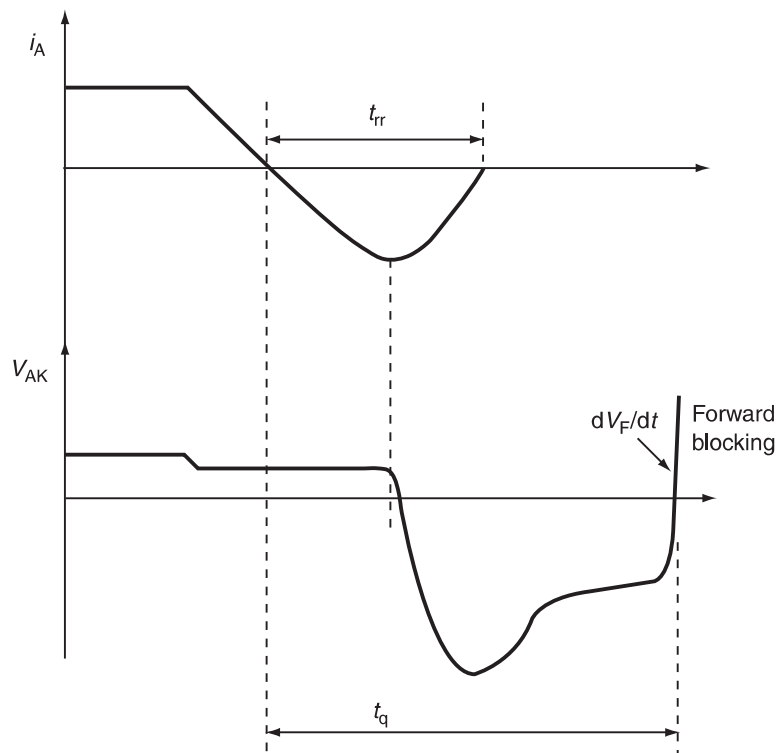


Figure 1.41 | Anode current and anode–cathode voltage during turn-OFF.

is larger than the latching current of the device, then the device will self-trigger and go into conduction even in the absence of the gate current. Therefore a high dv/dt has the possibility of undesirable latch up of the device. Practical SCRs have dv/dt limits of about 200–500 V/ μ s. In circuits wherein the dv/dt exceeds the rated dv/dt limits of the device, additional circuits called snubber circuits will have to be used to lower the dv/dt .

Other Parameters

The various current ratings like I_{Tav} , I_{Trms} , I_{Tsurge} are similar to those discussed for the diode. The ratings are a bit complicated compared to a diode. The value given in the datasheet for the mean ON-state current I_{Tav} is valid for a certain waveform and case temperature. I_{Trms} signifies the heating effect due to i^2R dissipation and is limited due to the thermal stress on the device. I_{TSM} is the maximum permissible peak current of half sine wave with a duration of 10 ms at a specified temperature.

Power dissipated in a thyristor is calculated in the same way as that of the diode. Gate-current losses add to the power losses in the case of thyristors as compared with the diode losses. Power loss calculation depends on the type of waveform too. Thyristors are very rugged devices as they can carry considerable overload currents without exceeding the junction temperature.

Circuit Model

There are many circuit models of SCRs available in the literature and circuit simulator libraries like SPICE. One such simple model is given here. It can be constructed using the two-transistor analog of an SCR as shown in Figure 1.42. Diode DFOR is added to model the forward breakdown. A resistance is added across the base-emitter of Q_2 to provide a discharge path for the stored charges in Q_2 . The various parameters for the PNP and NPN transistors are included in the model to give a satisfactory simulation performance.

DIACs and TRIACs

DIAC is a two-terminal device without any gate. It is a bi-directional switch. The breakdown voltage is generally kept low (~ 30 V) because it is generally used as a threshold device in many applications. Figure 1.43(a) shows the DIAC characteristics and Figure 1.43(b) depicts its symbol.

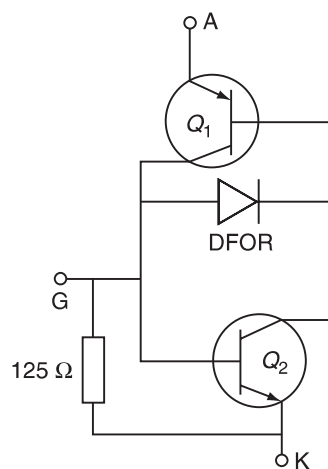


Figure 1.42 | Circuit model of an SCR.

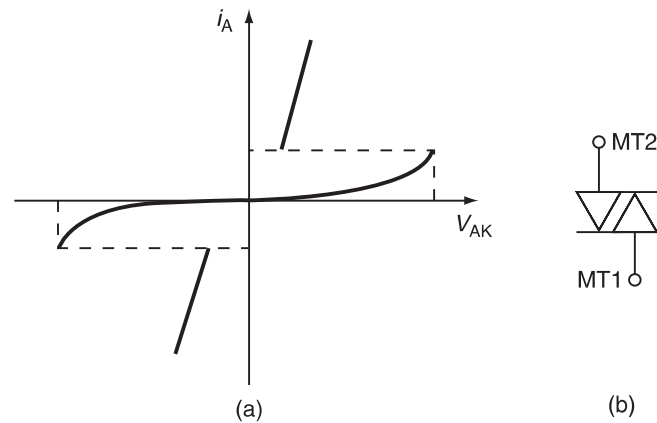


Figure 1.43 | DIAC: (a) $v-i$ characteristic; (b) symbol.

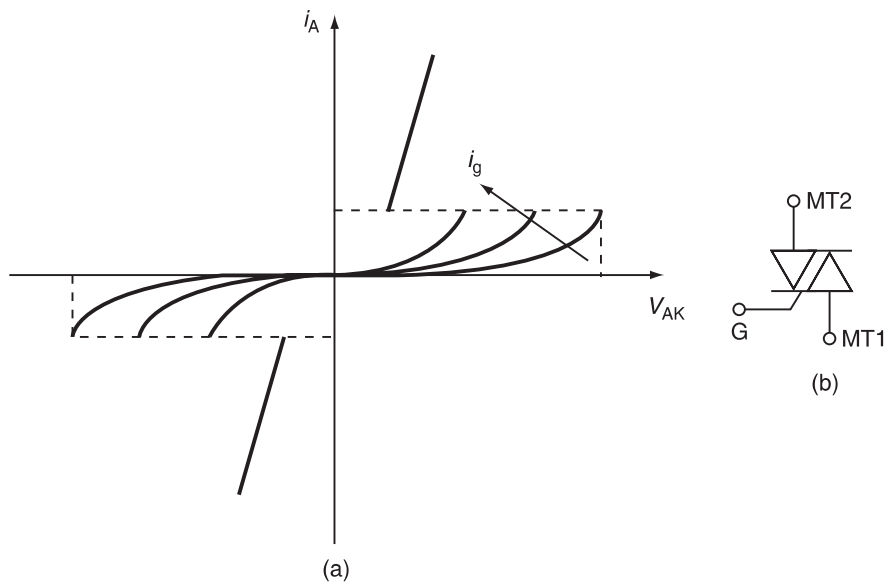


Figure 1.44 | TRIAC: (a) $v-i$ characteristic; (b) symbol.

TRIAC is also a bi-directional thyristor switch but it has a gate terminal that is used for controlling the turn-ON of the TRIAC. It is mainly used for AC control applications. As the switch is bi-directional, its terminals are called main terminal 1 and 2 (MT1 and MT2), instead of anode and cathode. Figure 1.44(a) shows the characteristics of the TRIAC and Figure 1.44(b) depicts its symbol. A simple TRIAC control circuit is shown in Figure 1.45. This circuit uses a DIAC as a threshold device for triggering the TRIAC at a particular source phase angle.

There are different ways in which the gate can be fired. The most common is with reference to MT1. The polarity of voltage across MT2 and MT1 decides the direction of current flow, and gate pulse is always positive irrespective of direction of current flow. TRIACs are generally used for 50 Hz applications. With inductive load, the TRIAC will be subjected to large dv/dt and therefore protective snubber circuits must be used.

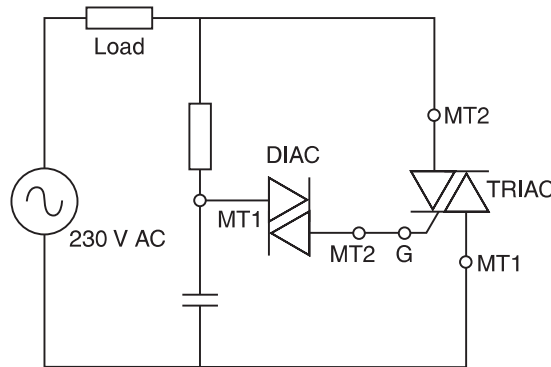


Figure 1.45 | TRIAC control circuit using DIAC.

Gate Turn-OFF Thyristor (GTO)

It can be recalled that an SCR cannot be turned OFF by gate control. It needs an extra turn-OFF/commutation circuit. GTO has been devised to provide turn-OFF through the gate terminal. The GTO is a four-layer device that can be turned ON with a positive gate current pulse and turned OFF with a negative gate current pulse. Figure 1.46(a) shows some of the commonly used GTO symbols.

The physical structure of a GTO is shown in Figure 1.46(b). Referring to figure, it may be observed that the gate and cathode are interspersed. Such a sectionalized geometry helps in having better gate control. The anode layer has n - and p -regions interspersed as shown in Figure 1.46(b). This gives a structure termed as the “*anode short*” structure that is found to improve the turn-OFF time.

GTO action can also be described by the two-transistor model as discussed for the case of an SCR. By suitably controlling the doping profiles of the n - and p -regions, the α and β of the transistors, it is possible to achieve gate turn-OFF with a negative gate current. Removal of excess charge is an important part of the turn-OFF process. The negative gate current and the anode current during the turn-OFF process are depicted in Figure 1.47.

The SCR and GTO differences are highlighted below:

1. GTO has higher ON-state drop as compared to the SCR. The GTO has an ON-state drop of about 3–4 V whereas the SCR has an ON-state drop of around 1.5–2 V.
2. The latching and holding currents are high for a GTO. As an example, a typical 110 A SCR has I_H of 150 mA and I_L of 300 mA, whereas even a low current GTO, a typical 18 A GTO has $I_H = 600$ mA and $I_L = 900$ mA. For 30 A GTO, they are more than 1 A. Therefore the triggering pulse must be long enough or continuous for the GTO unlike the case of the SCR.
3. The reverse-blocking capability of a GTO is very low (~ 20 V) and hence cannot be used in rectifier applications.
4. When the device is to be turned OFF, the turn-OFF amplification factor (anode current with respect to the turn-OFF gate current) can be anywhere between 1 and 5 and the turn-OFF time between 1 and several milliseconds. There is a maximum current rating beyond which gate cannot turn OFF the GTO. This is called repetitive controllible ON-state current (or extinguishable current).
5. GTO has a large dv/dt capability (1000 V/ μ s) compared to a SCR (200–500 V/ μ s).
6. GTOs can be used up to 1–4 kHz of switching frequency range.

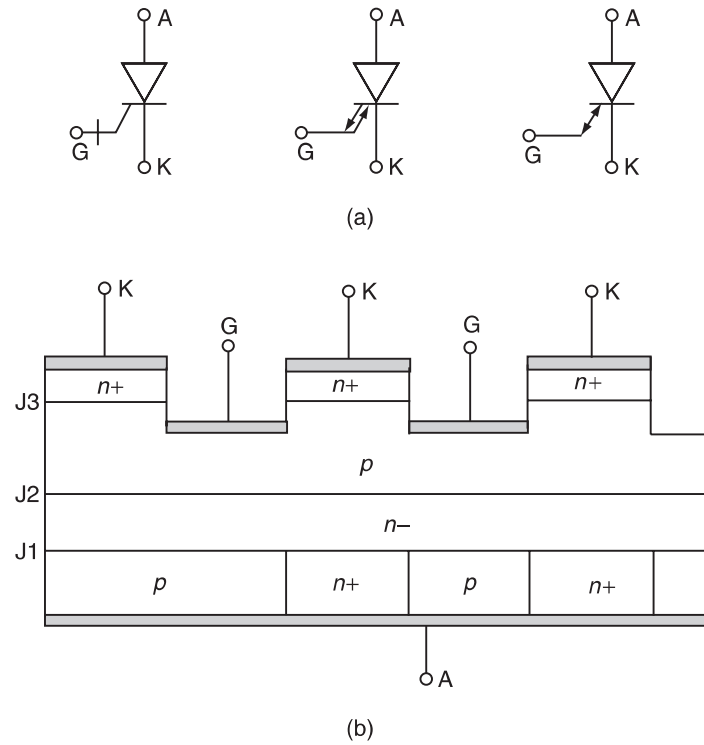


Figure 1.46 | (a) Symbols of GTO; (b) structure of a GTO.

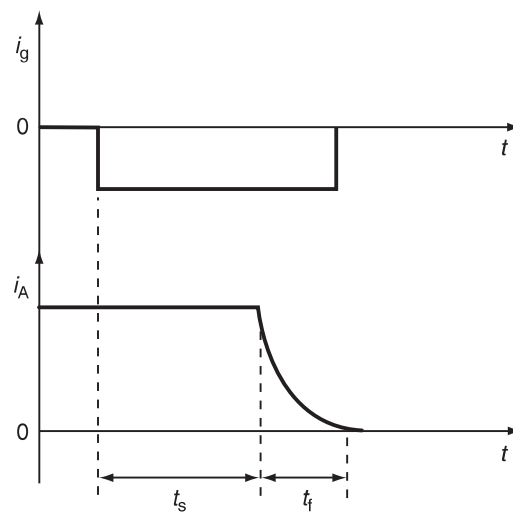


Figure 1.47 | Turn-OFF characteristics of a GTO.

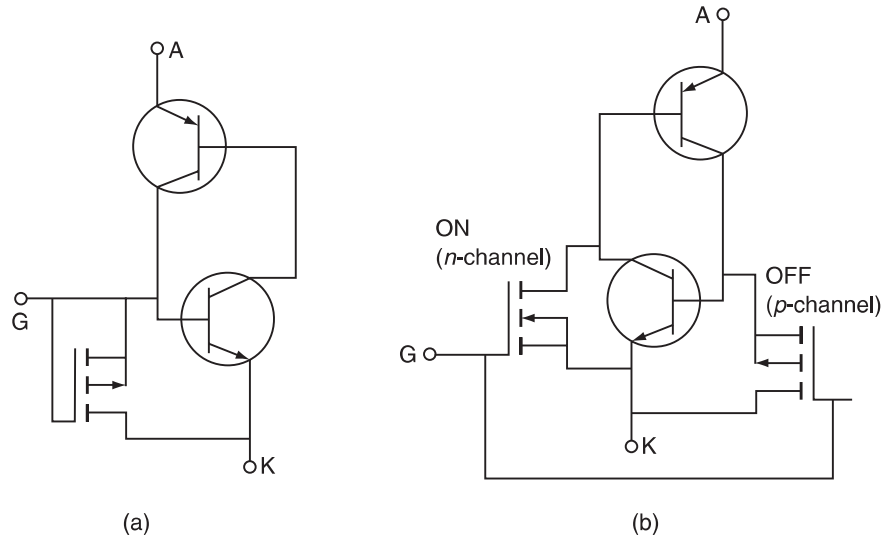


Figure 1.48 | MCT configurations.

MOS-Controlled Thyristor (MCT)

MCT is an interesting device wherein both the turn-ON and turn-OFF can be controlled. However, commercially it is not as popular as the MOSFETs and the IGBTs. Figure 1.48 shows some possible MCT functional configurations. Figure 1.48(a) shows an MCT where turn-ON is done in the usual way but turn-OFF is done using a MOSFET. Figure 1.49(b) shows another configuration wherein an n -channel MOSFET is used for turning ON the MCT and a p -channel MOSFET is used for turning OFF the MCT. The major features of an MCT are:

1. It has low conduction voltage drop: $1/3$ to $1/2$ of a comparable IGBT.
2. It combines high-speed, low-power gate turn-ON–turn-OFF advantages.
3. It has high dv/dt and di/dt capabilities.

CONCLUDING REMARKS

This chapter discussed few important viewpoints of the semiconductor devices giving a flavor for the operation of the various devices as a power switch. It should, however, be observed that no single switch meets all the features of an ideal switch as described in Section 1.1. One should be aware of the non-idealities of each type of the power switch so that the power electronic circuits can be modeled with greater closeness to the physical system.

There are many semiconductor devices available commercially that have been fabricated with newer and improved technologies. However, this chapter discusses only a few generic types, highlighting the important issues that need to be studied and understood when encountered with a new device. With special regard to the semiconductor devices being used as power switches in power electronic circuits, one should study with emphasis on

three aspects in general, viz. (a) static characteristics that give insight on the steady-state operating points, (b) dynamic characteristics that give the behavior of the switch during turn-ON and turn-OFF and (c) power loss within the device due to conduction and switching.

The theoretical insights should be cemented with the experience that can be gained from practical work to strengthen the understanding of the

device. The next section provides some laboratory exercises that should be implemented both on a simulation platform like spice and by hardware breadboarding to appreciate the functional features of a particular power semiconductor switch. The laboratory exercises are focused on popular devices like diodes, BJTs, MOSFETs and IGBTs. However, the exercises can be extended for any other power semiconductor switch too.

LABORATORY EXERCISES

1. Consider the diode test circuit shown in Figure 1.49 where V_i is a voltage source and D is a diode. The diode D is appropriately chosen such that the current and voltages are within ratings.

Mode of implementation: The above circuit can be studied by

- a. Simulation in Spice
- b. Hardware breadboarding

Tasks for study:

- (a) Use a variable DC voltage source. Measure the current through and the voltage across the diode for different input DC voltage values. Reverse the polarity of the input voltage source and measure the current through and the voltage across the diode for different input voltage values. Tabulate the diode current and voltage values and plot the i - v characteristics of the diode.

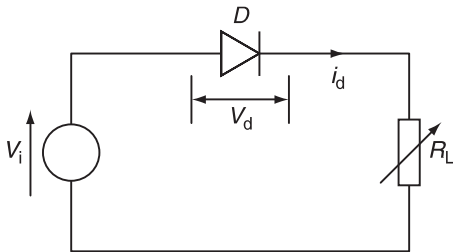


Figure 1.49 | Diode test circuit.

- (b) Find out the forward dynamic resistance of the diode from the i - v characteristic.
 - (c) Use a square waveform input voltage source and observe the current through the diode and the voltage across it during turn-ON and turn-OFF.
 - (d) Compute the experimental switching loss and compare with the theoretical estimate.
 - (e) Compute the experimental conduction loss and compare with the theoretical estimate.
 - (f) Repeat steps (a)–(c) for different loads by varying R_L .
 - (g) Repeat steps (a)–(c) for different source frequencies. What is the effect on the diode power loss?
 - (h) Measure the t_{rr} of the diode from the current waveform through the diode. What is the effect of the load on t_{rr} ?
 - (i) Measure the reverse recovery charge Q_{rr} from the experimental results.
 - (j) Use a sinusoidal waveform input voltage source and observe the current through and the voltage across it during voltage transition from positive to negative. Compute the power loss in the device and compare the loss with that obtained when using a square waveform input source.
2. Consider the two BJT test circuits shown in the Figure 1.50 where V_b is the base drive pulse source. V_b should be chosen such that it is capable

of both sourcing and sinking current. V_c is the collector DC supply voltage. R_c is the collector load and R_b is the base drive resistor. The BJT is chosen with appropriate rating to handle the collector current and the maximum V_{ce} as decided by R_c and V_c respectively. The circuit of Figure 1.50(a) consists of a resistive collector load and that of Figure 1.50(b) consists of an R – L load. The diode D is connected as shown to provide a freewheeling path for the inductor current when the BJT Q is turned OFF.

Mode of implementation: The above circuit can be studied by

- a. Simulation in Spice
- b. Hardware breadboarding

Tasks for study:

- (a) Set V_b to be a DC source. Measure the collector current and the collector–emitter voltage across transistor for different values of R_b . Tabulate the BJT collector current, base current, emitter current and collector–emitter voltage values. Find the

α and β parameters. Plot the I_c – V_{ce} characteristics for different base current values.

- (b) Set V_b to be a pulse source of frequency 20 kHz. Measure the collector current and the collector–emitter voltage waveforms and observe the turn-OFF and turn-ON portions.
 - (c) Compute the experimental switching loss and compare with the theoretical estimate.
 - (d) Compute the experimental conduction loss and compare with the theoretical estimate.
 - (e) Repeat steps (b)–(d) for different loads by varying R_L .
 - (f) Repeat steps (b)–(d) for different base pulse frequencies. What is the effect on the BJT power loss?
 - (g) What is the effect of R_b on the turn-ON and turn-OFF times? Why?
 - (h) Perform the steps (b)–(g) for the inductive load test circuit of Figure 1.50(b).
3. Consider the two MOSFET test circuits shown in Figure 1.51 where V_g is the gate drive pulse

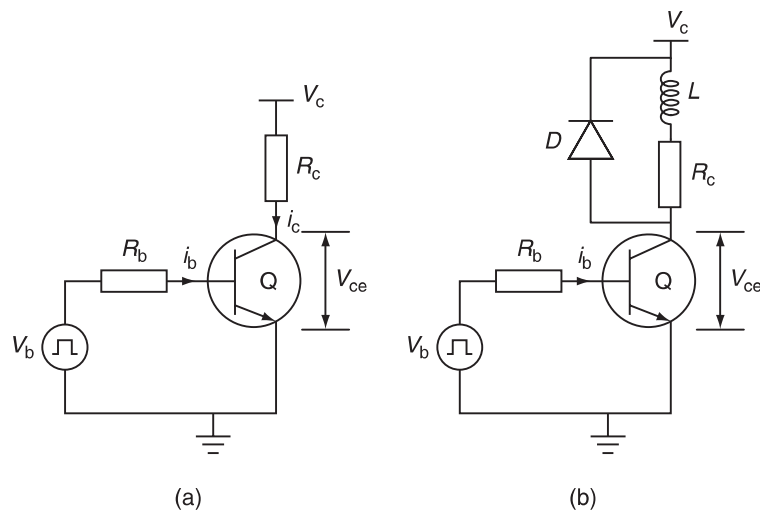


Figure 1.50 | BJT test circuit.

source. V_g should be chosen such that it is capable of both sourcing and sinking current. V_d is the drain DC supply voltage. R_d is the drain load and R_g is the gate drive resistor. The MOSFET is chosen with appropriate rating to handle the drain current and the maximum V_{ds} as decided by R_d and V_d , respectively. The circuit of Figure 1.51(a) consists of a resistive drain load and that of Figure 1.51(b) consists of an R – L load. The diode D is connected as shown to provide a freewheeling path for the inductor current when the MOSFET is turned OFF.

Mode of implementation: The above circuit can be studied by

- Simulation in Spice
- Hardware breadboarding

Tasks for study:

- Set V_g to be a DC source. Measure the drain current and the drain–source voltage across the MOSFET for different values of V_g . Tabulate the MOSFET drain current, and drain–source voltage values. Plot the

I_d – V_{ds} characteristics for different gate voltage values.

- Set V_g to be a pulse source of frequency 20 kHz. Measure the drain current and the drain–source voltage waveforms and observe the turn-OFF and turn-ON portions.
- Compute the experimental switching loss and compare with the theoretical estimate.
- Compute the experimental conduction loss and compare with the theoretical estimate.
- Repeat steps (b)–(d) for different loads by varying R_L .
- Repeat steps (b)–(d) for different gate pulse frequencies. What is the effect on the MOSFET power loss?
- What is the effect of V_g on the turn-ON and turn-OFF times?
- What is the effect of R_g on the turn-ON and turn-OFF times?
- Measure the gate current and explain its pulse shape.

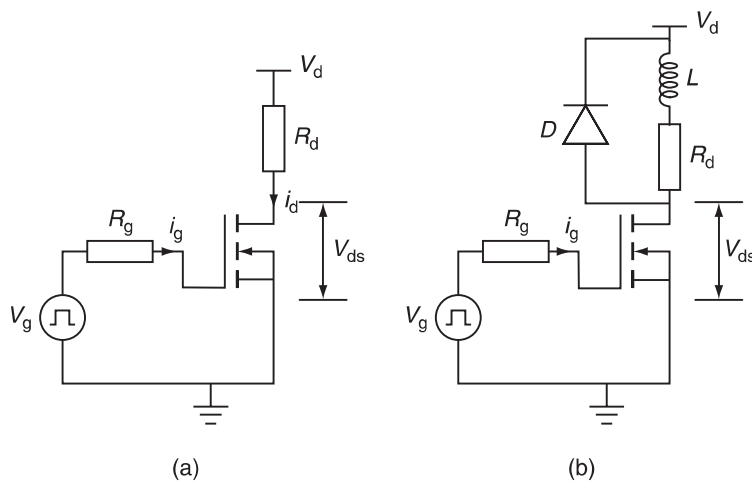


Figure 1.51 MOSFET test circuit.

- (j) Perform the steps (b)–(g) for the inductive load test circuit of Figure 1.51(b).
 - (k) Compare the switching speed and power loss of MOSFET with comparable BJT.
4. Consider the two IGBT test circuits shown in Figure 1.52 where V_g is the gate drive pulse source. V_g should be chosen such that it is capable of both sourcing and sinking current. V_c is the collector DC supply voltage. R_c is the collector load and R_g is the gate drive resistor. The IGBT is chosen with appropriate rating to handle the collector current and the maximum V_{ce} as decided by R_c and V_c , respectively. The circuit of Figure 1.52(a) consists of a resistive collector load and that of Figure 1.52(b) consists of an R – L load. The diode D is connected as shown to provide a freewheeling path for the inductor current when the IGBT is turned OFF.

Mode of implementation: The above circuit can be studied by

- a. Simulation in Spice
- b. Hardware breadboarding

Tasks for study:

- (a) Set V_g to be a DC source. Measure the collector current and the collector–emitter voltage across transistor for different values of V_g . Tabulate the IGBT collector current and collector–emitter voltage values. Plot the I_c – V_{ce} characteristics for different gate voltage values.
- (b) Set V_g to be a pulse source of frequency 20 kHz. Measure the collector current and the collector–emitter voltage waveforms and observe the turn-OFF and turn-ON portions.
- (c) Compute the experimental switching loss and compare with the theoretical estimate.
- (d) Compute the experimental conduction loss and compare with the theoretical estimate.
- (e) Repeat steps (b)–(d) for different loads by varying R_L .
- (f) Repeat steps (b)–(d) for different gate pulse frequencies.
- (g) Perform the steps (b)–(f) for the inductive load test circuit of Figure 1.52(b).

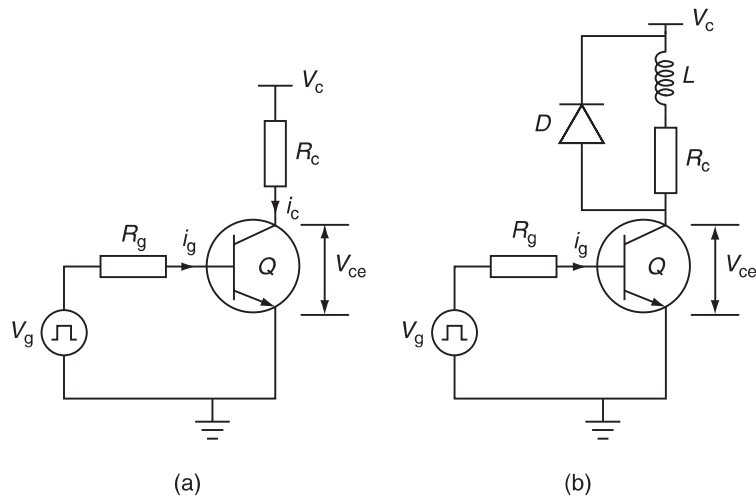


Figure 1.52 | IGBT test circuit.

FILL IN THE BLANKS

1. Diodes are devices that have two terminals: _____ and _____.
2. When the anode is more positive than the _____, the diode is said to be forward-biased.
3. When the _____ is more positive than the _____, the diode is said to be reverse-biased.
4. During turn-OFF of the diode, the _____ stored in the diffusion region has to be _____ before the junction can be reverse-biased.
5. As long as there are excess charge carriers in the diffusion region, the junction will be in _____.
6. The bipolar transistor is a _____ device.
7. BJT is a _____ device.
8. BJT is a _____ device.
9. Based on the doping type, there are two types of BJT: _____ and _____.
10. Larger the collector region, _____ is the amount of voltage that the transistor can withstand during the OFF-state.
11. The ratio of the collector current to the emitter current is called the _____ of the transistor.
12. The alpha parameter varies between _____ and _____.
13. Using the Eber-Molls model, the collector current is given as _____.
14. Using the Eber-Molls model, the emitter current is given as _____.
15. The ratio of the collector current to the base current is given by _____.
16. The electric field across the transistor is dependent on the _____ and the _____.
17. The electric field increases with _____ and with _____.
18. The safe operating areas give information about the capability of a device to handle the specified _____.
19. MOSFET is a _____.
20. The Ohmic region resistance $R_{DS(ON)}$ is due to _____.
21. $R_{DS(ON)}$ has a _____ temperature coefficient.
22. Compared to MOSFETs, BJTs have _____ conduction losses but _____ switching-losses.
23. BJTs are _____-controlled devices whereas MOSFETs are voltage-controlled devices.
24. IGBT is a _____.
25. IGBT can support _____ voltages during the OFF-state.
26. In comparison, for a given i_c , the $V_{ce(sat)}$ of the IGBT is _____ than the V_{ds} of a comparable MOSFET when ON but _____ than $V_{ce(sat)}$ of a comparable BJT.
27. IGBT has _____ overload capability as compared to a MOSFET.
28. IGBT has _____ overvoltage robustness than MOSFETs.
29. IGBTs are more like the _____ during turn-OFF and like the _____ during turn-ON.
30. Thyristors is a generic name for a semiconductor switch having _____ layers.
31. Application of a _____ voltage to the anode-cathode of an SCR will reverse-bias the SCR.
32. DIAC and TRIAC are _____ switches.
33. In a GTO, the turn-OFF is _____ whereas in an SCR, the turn-OFF is _____.

34. GTO has _____ ON-state drop as compared to the SCR.
35. The latching and holding currents are _____ for a GTO as compared to the SCR.
36. The reverse-blocking capability of a GTO is very low and hence cannot be used in _____ applications.

DESCRIPTIVE QUESTIONS

1. What are the characteristic features of an ideal switch?
2. What is a controlled switch?
3. What is an uncontrolled switch?
4. What is the depletion region?
5. *"The value of the potential barrier in volts depends on the charge carriers and is governed by the Boltzmann's relation."* Discuss.
6. With respect to the diode's V - I static characteristics, explain the difference among the ideal, piece-wise linear and the actual V - I characteristics.
7. Discuss the turn-OFF process in the diode.
8. What is meant by reverse recovery time for a diode?
9. What is the ON-state loss in a diode?
10. What is the switching loss in a diode?
11. Describe the switching process within a transistor.
12. Derive relationship for the power dissipation within the transistor between the applied collector-emitter voltage, the collector-current and the switching frequency of the transistor for (a) resistive load in the collector, (b) resistive-inductive load in the collector and c) resistive-capacitive load across the collector-emitter of the transistor.
13. Discuss forward and reverse safe operating areas.
14. What is second breakdown?
15. How are transistors connected in parallel? Discuss.
16. How does the current gain, or the beta parameter, increase with the Darlington configuration?
17. What are the semiconductor layers in a MOSFET structure?
18. What is the inversion layer? How is it formed?
19. Discuss the MOSFET structure and the various associated capacitances?
20. How are the MOSFET capacitances C_{iss} , C_{oss} and C_{rss} related to the parasitic capacitances C_{gs} , C_{gd} and C_{ds} ?
21. What is gate threshold voltage?
22. What is Ohmic region of the MOSFET static characteristics?
23. Explain the MOSFET turn-OFF and turn-ON processes?
24. How is the positive temperature coefficient advantageous for paralleling of MOSFETs?
25. Discuss the functional structure of the IGBT.
26. Explain the turn-ON and turn-OFF of IGBT with waveforms.
27. What is tail current in IGBT?
28. What are holding and latching currents in SCR?
29. What is forward breakover voltage in SCR?
30. Discuss the effect of di/dt and dv/dt on the SCR switch.

PROBLEMS

- Find the ratio of the diffusion current to the saturation current for a p – n junction at 75°C having a forward potential of 0.6 V applied across it.
- A diode and a $10\ \Omega$ resistor are connected in series to a square wave voltage source of 50 V peak. Find the conduction loss for the diode if the forward dynamic resistance is (a) $0.1\ \Omega$ and (b) $0.2\ \Omega$ at the operating point. (Assume the forward barrier potential to be 0.7 V .)
- A diode and a $10\ \Omega$ resistor are connected in series to a pulse voltage source of 50 V peak. If the forward dynamic resistance is $0.1\ \Omega$ at the operating point, then find the conduction loss of the diode when (a) pulse frequency is 20 kHz and pulse width $40\ \mu\text{s}$, (b) pulse frequency is 20 kHz and pulse width is $10\ \mu\text{s}$, (c) pulse frequency is 10 kHz and pulse width is $80\ \mu\text{s}$ and (d) pulse frequency is 10 kHz and pulse width is $20\ \mu\text{s}$.
- A diode and a $10\ \Omega$ resistor are connected in series to a square wave voltage source of 50 V peak. The reverse recovery time for the diode is given to be 200 ns . Find the switching loss of the diode when (a) input frequency is 100 kHz and (b) input frequency is 50 kHz .
- For Problem 4 above, estimate the reverse recovery charge.
- A BJT is driving a 10 A resistive load from a 100 V DC supply. The base drive signal is switching at frequency of 50 kHz and duty cycle of 0.75 . The BJT has the following datasheet specifications: $V_{\text{besat}} = 0.7\text{ V}$, $V_{\text{ceo}} = 30\text{ V}$, $I_{\text{cm}} = 15\text{ A}$, $V_{\text{cesat}} = 0.3\text{ V}$, $h_{\text{FEmin}} = 100$, $t_{\text{d}} = 1\ \mu\text{s}$, $t_{\text{s}} = 2\ \mu\text{s}$, $t_{\text{r}} = 1.5\ \mu\text{s}$, $t_{\text{f}} = 1.5\ \mu\text{s}$. Calculate the power loss in the BJT.
- For Problem 6 above, plot the device power dissipation as the switching frequency varies from 1 kHz to 100 kHz .
- Two BJTs are connected in parallel to share the load current. In order to ensure sharing, two equal-valued resistors with value R are connected in the emitter leads of the BJTs. For a mismatch in the base–emitter voltage of 0.2 V between the two BJTs, a mismatch in the collector currents of the two BJTs should be less than 1 A . Calculate the value of the resistors that needs to be connected in the emitter leads.
- A MOSFET is operated such that the operating point is in the active region. The MOSFET has a gate–source threshold voltage value of 2.5 V . A gate–source voltage of 5 V is applied to the gate–source terminals of the MOSFET which results in the flow of drain current. On increasing the gate–source voltage to 7.5 V , what is the factor by which the drain current increases?
- A MOSFET is driving a 10 A resistive load from a 100 V DC supply. The base drive signal is switching at frequency of 100 kHz and duty cycle of 0.6 . The MOSFET has the following datasheet specifications: $R_{\text{DS(ON)}} = 0.1\ \Omega$, $t_{\text{r}} = 100\text{ ns}$, $t_{\text{f}} = 150\text{ ns}$. Calculate the conduction and switching power losses in the MOSFET.

ANSWERS

Fill in the Blanks

- | | | |
|----------------------|---------------------------|-------------------------|
| 1. anode and cathode | 3. cathode; anode | 5. forward-biased state |
| 2. cathode | 4. excess charge; removed | 6. three-terminal |

- | | | |
|--|-----------------------------------|------------------------------|
| 7. current-controlled | 17. increasing collector voltage; | 26. less; greater |
| 8. dual junction | increasing collector-current | 27. greater |
| 9. NPN transistor and PNP | density | 28. less |
| transistor | 18. current and voltage | 29. BJTs; MOSFETs |
| 10. greater | 19. voltage-controlled device | 30. four or more |
| 11. alpha parameter | 20. majority carriers | 31. negative |
| 12. 0.9 and 0.98 | 21. positive | 32. bi-directional |
| 13. $I_c = \alpha_F I_F - I_{co}(e^{V_c/V_T} - 1)$ | 22. lower | 33. controlled; uncontrolled |
| 14. $I_e = \alpha_R I_R - I_{co}(e^{V_e/V_T} - 1)$ | 23. current | 34. higher |
| 15. β or h_{FE} | 24. voltage-controlled device | 35. higher |
| 16. collector-current density; | 25. bipolar | 36. rectifier |
| applied collector-emitter | | |
| voltage | | |